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An interactive decision making framework design for the outsourcing cooperation between the service provider and the airline: An exact bilevel method

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Highlights

- Design an interactive decision making framework to achieve win–win cooperation between the MSP and the airline.
- Propose a novel exact algorithm to solve the bilevel MIP model.
- Solve the bilevel MIP model to optimality or near-optimality within reasonable time.
- Validate the superiority of the proposed bilevel model.
- Identify key factors influencing Pareto improvements through sensitivity analysis.

Abstract

In reality, more and more airlines outsource maintenance tasks to the maintenance, repair, and overhaul (MRO) service provider to achieve cost-effective operational planning. However, the existing literature mainly focuses on the optimal decisions from the perspective of the airline, which usually ignores the resource scarcity of the maintenance service provider (MSP) and provides “virtual optimal” solutions for aircraft maintenance routing problem (AMRP). In actual operations, the MSP may not be able to provide “mutually beneficial” maintenance services to its customers due to its resource scarcity and the lack of information sharing. Therefore, in this paper, based on industry practices we model an aviation service supply chain consisting of one MSP and one airline, in which the MSP may provide maintenance services to multiple airlines with limited spatiotemporal resources

with different prices. A bilevel mixed integer programming (MIP) model is formulated to characterize the interactive decision-making structure between the MSP and the airline that aims to help increase the operational efficiency of both the MSP and the airline. An exact bilevel solution framework is developed which is verified to achieve convergence and obtain the bilevel optimality. Through the proposed algorithm, the computational study is further conducted based on the actual operational data from one of the biggest Chinese airline companies. The results verify that, compared to the benchmark, the proposed bilevel model can achieve Pareto improvement. In addition, the sensitivity analyses regarding pricing, flexible maintenance resources and opportunity cost coefficients verify the effectiveness and robustness of the proposed bilevel model, and further derive some useful managerial insights for both the MSP and the airline at strategic and operational levels.



Keywords

Bilevel mixed integer programming; Aircraft maintenance routing; MRO outsourcing; Column generation

1. Introduction

1.1. Background and Motivation

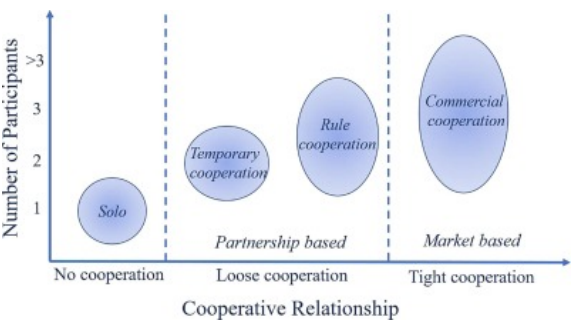
In the current global [aviation industry](#), outsourcing aircraft maintenance has become a widespread practice ([Qin et al., 2024](#)). Statistically, the global maintenance, repair, and overhaul (MRO) market grew by 18% in 2022 and is projected to expand further, reaching \$139 billion over the next decade.¹ According to FAA [U.S. Civil Airman Statistics](#), global demand for new maintenance technicians is expected to increase by approximately 50% over the next twenty years. Aircraft maintenance operations are strictly regulated, requiring that maintenance tasks be performed by qualified technicians holding specific licenses for particular aircraft models. Consequently, smaller or low-cost [airlines](#), which often lack extensive in-house maintenance capabilities, prefer outsourcing maintenance tasks. This allows them to concentrate on their core operations while ensuring professional and safe aircraft maintenance. Even major airlines frequently outsource certain maintenance tasks, particularly those involving specific aircraft models or those requiring qualifications unavailable at their own facilities.

However, the MRO sector currently faces critical challenges, including significant shortages of qualified technicians and escalating operational costs stemming from post-pandemic fiscal policies. The former chairman of the NBAA Maintenance Committee highlighted that for every four aviation maintenance technicians leaving the industry, only three new technicians enter the workforce.² Oliver Wyman reported a notable shortage of 24,000 airline maintenance technicians in North America alone.³ This shortage poses considerable challenges for maintenance service providers (MSPs), impacting their ability to maintain timely service performance amidst increasing operational costs.

The outsourcing of maintenance services has been scrutinized since the 1990s, particularly regarding its impact on aviation safety. [Bağcı and Gereke \(2019\)](#) argued that outsourcing practices impose significant cost and time pressures on MSPs, complicating the allocation of technicians and potentially compromising safety. [Nam et al. \(2023\)](#) similarly observed that cost-driven outsourcing by airlines often results in technician shortages at outsourced maintenance stations, indirectly affecting flight safety. Hence, efficiently utilizing maintenance resources is a critical challenge for MSPs, especially under conditions of resource scarcity. MSPs often prioritize maintenance needs based on strategic partnerships or major customer relationships, potentially creating a mismatch between airlines' maintenance demands and available resources.

Airlines can employ various cooperation models for aircraft maintenance ([Kilpi et al., 2009](#)). As shown in [Fig. 1](#), the traditional "Solo" strategy involves airlines independently managing their maintenance tasks. Alternatively, temporary cooperation through informal resource borrowing can reduce reserve costs but leads to unstable resource supplies. Formalized "rule cooperation" agreements ensure more stable resource availability and enhanced utilization through official contractual arrangements. Additionally, commercial cooperation between airlines and MSPs involves outsourcing availability services. Under this arrangement, airlines pay a fixed annual fee to access spare resources from MSPs when needed. Such cooperation requires detailed agreements stipulating service fees, delivery timelines, and delay-related responsibilities.





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Fig. 1. Different Modes of Cooperation in Aviation Maintenance.

In this study, we focus explicitly on commercial cooperation between airlines and MSPs. Given that independent MSPs typically serve multiple airlines simultaneously (see Table 1), resource availability for individual airlines cannot always be guaranteed. Addressing this critical challenge, we propose a dynamic and collaborative decision-making framework designed to optimize maintenance resource utilization while balancing the interests of both airlines and MSPs. Previous studies have inadequately addressed the complex interactions between airlines and MSPs, often overlooking mutual influences and constraints inherent in real-world decision-making processes (Saltzman and Stern, 2022, Zhang et al., 2024). Such oversights result in theoretically optimal solutions difficult to implement practically. Unlike prior research, our study explicitly incorporates an interactive decision making process between airlines and MSPs within the context of resource scarcity. Rather than assuming unlimited MSP resources or treating resource constraints in isolation, we present an interactive decision-making framework that accounts for both parties' constraints and interests, fostering mutual benefits through flexible resource coordination involving maintenance technicians and spatiotemporal resources.

Table 1. Different types of MSPs and their partial customers.

Types of MSPs	Some MSPs Directory	Customers in Partial
Airline-Owned Maintenance Services Subsidiaries or Joint Ventures of Global Maintenance Services Giants	AMECO, ^a GAMECO, ^b STARCO, ^c TAECO, ^d	AMECO: Asiana Airlines, China National Aviation, Air China, Shenzhen Airlines, China Southern Airlines, SF Airlines; GAMECO: China Southern Airlines, FedEx, Asiana Airlines, SF Airlines, Lufthansa, Hangzhou YTO, Postal Service, Icelandair, Philippine Airlines, Aeromexico, Chongqing Airlines, Japan Airlines; STARCO: SpiceJet (India), All Nippon Airways (Japan), Air Bridge (Russia), Air Macau; TAECO: Japan Airlines, Air Canada, Finnair, Luxembourg Cargo Airlines, Changlong Airlines, Zhongzhou Airlines, China Airlines, Emirates.
Independent Third-Party Maintenance Providers	SR Technics, ^e Lufthansa Technik, ^f Hangxin Technology, ^g Hitec, ^h Andawell, ⁱ Wuhan Hangda ^j	SR Technics: Pacific Airlines, Asiana Airlines; Lufthansa Technik: Swiss Airlines, easyJet, VietJet Air, Philippine Airlines; Hangxin Technology: XIAMEN AIR, MIAT Mongolian Airlines, Scandinavian Airlines, Austrian Airlines, Finnair, Hainan Airlines, CapitalAirlines; Hitec: Air China, China Southern Airlines, China Eastern Airlines; Andawell: China Eastern Airlines, SF Airlines, Shandong Airlines, State Grid General Aviation, China General Aviation; Wuhan Hangda: Air China, China Southern Airlines, China Eastern Airlines, Xiamen Airlines, Shandong Airlines, Hong Kong Airlines.

1
<https://www.ameco.com.cn/cn/news/index/4> ↗ (Accessed on Dec 2024).

2
<https://www.gameco.com.cn/> ↗ (Accessed on Dec 2024).

3

<https://www.carnoc.com/subject/starco/> ↗ (Accessed on Dec 2024).

4

<https://www.haeco.com/sc/media-centre/press-releases> ↗ (Accessed on Dec 2024).

5

<https://www.srtechnics.com/> ↗ (Accessed on Dec 2024).

6

<https://www.lufthansa-technik.com/en/press-releases> ↗ (Accessed on Dec 2024).

7

https://hangxin.com/cn/qygk/index_3.aspx ↗ (Accessed on Dec 2024).

8

<https://www.schtgx.com/> ↗ (Accessed on Dec 2024).

9

https://www.andawell.com/aboutus/client/china_airlines/ ↗ (Accessed on Dec 2024).

10

<https://www.hangdagroup.com/Customerbase> ↗ (Accessed on Dec 2024).

Specifically, we propose a novel bilevel model designed to optimize spatiotemporal resource scheduling and coordination between airlines and MSPs. To better align with real-world scenarios, we incorporate the concept of “flexible maintenance resources” into our model. These resources, such as temporarily hired technicians,⁴ collaborative arrangements with other organizations (e.g., Spirit AeroSystems and Malaysia Airlines),⁵ and emergency reserve resources, typically incur higher costs but provide essential flexibility. In our proposed model, airlines have the option to utilize these flexible resources at premium costs, subject to MSP availability. This iterative approach continues until optimal decisions are reached, effectively mirroring realistic conditions of resource constraints faced by MSPs serving multiple airlines.

1.2. Contribution

In this study, the main contribution of this paper is summarized below:

We propose a bilevel mixed integer programming (MIP) model to capture the interactive decision-making process between the MSP and the airline company. Unlike conventional models that assume the airline as the “leader,” our model positions the MSP as the leader due to resource scarcity and its diverse client base. This novel approach better reflects the operational realities in aircraft maintenance decision-making. Our model explicitly incorporates critical economic aspects of both stakeholders, including maintenance costs, extra payment for flexible maintenance resources, opportunity costs, and flight operating costs. Practical constraints, such as limits on flight hours, takeoffs, and technician requirements, and regulations governing maintenance schedules are also integrated into the framework.

We solve the bilevel model via a novel exact algorithm framework called the greedy search nested column generation algorithm (GSNCG). Due to the characteristics and complexity of the proposed bilevel model, the existing exact methods for discrete bilevel models do not apply to our problem. In this paper, we propose a nested algorithm, in which the upper-level model is relaxed to obtain an upper bound for the bilevel model while the rational reaction from the lower-level model provides a lower bound. We rigorously prove that the algorithm can converge to get the optimal solution for the proposed bilevel MIP problem. Furthermore, we propose a column generation-based algorithm to solve the lower-level model to optimality or near-optimality. Specifically, due to the inherent complexity of the AMRP at the lower-level, three distinct sub-problems are defined according to the specific flight networks with different types of maintenance services. The optimality and efficiency of the solution approach are evaluated through extensive real-case-based computational experiments.

We validate our model through real-case computational experiments, demonstrating that the proposed interactive decision making framework yields a Pareto improvement for both the MSP and the airline compared to a sequential decision-making benchmark. Sensitivity analysis reveals conditions under which Pareto improvement may be achieved, highlighting the impact of pricing strategies, opportunity costs, and resource reservation levels on both parties' outcomes. Our findings indicate that proper pricing and opportunity cost strategies can promote a sustainable and mutually beneficial partnership between MSPs and airlines. This study is the first to evaluate such a quantitative cooperative mechanism at the operational level between these two entities, providing both theoretical and practical insights into aviation interactive decision-making framework design.

1.3. Structure of this study

The rest of this study is organized as follows. [Section 2](#) reviews the most related literature on aircraft maintenance routing planning, [resource allocation](#) (RA) by MSPs, and bilevel programming. [Section 3](#) presents the bilevel model for both RA by the MSP and the AMRP by the airline. [Section 4](#) details the proposed solution approach for the bilevel MIP model. [Section 5](#) introduces the [solution algorithm](#) for the lower-level AMRP model. [Section 6](#) describes the setup of the basic experiments and benchmark model. [Section 7](#) presents the results of the sensitivity analysis experiments. Finally, the [Section 8](#) summarizes this study.

2. Literature review

In this section, we review the relevant literature from three aspects. Firstly, we review the relevant research on AMRP and mainly discuss the main factors affecting aircraft maintenance routing planning. In addition, as our research also considers the [resource allocation](#) of the MRO, we review the relevant research on the resource allocation of the MRO. Finally, we review some related works on the solution approaches for bilevel programs.

2.1. Aircraft maintenance routing planning

Developing a complete [airline](#) plan is usually a process of four consecutive steps: flight scheduling, fleet assignment, aircraft routing, and crew scheduling ([Ahmed et al., 2022](#)). In general, these plans are addressed sequentially by developing a distinct model for each stage. Aircraft maintenance planning is typically integrated with the decision-making process for aircraft routing, known as aircraft maintenance routing planning (**AMRP**). [Sanchez et al. \(2020\)](#) pointed out that in a fiercely competitive environment, the objective of airlines is to minimize operational costs while ensuring service quality. Among these operational costs, a significant portion is attributed to maintenance. [Ma et al. \(2022\)](#) highlighted that AMRP is crucial for both passenger and cargo airlines to maintain operational efficiency and ensure that aircraft comply with [civil aviation](#) safety regulations regarding maintenance inspections.

AMRP determines the flight route for each aircraft to comply with FAA regulations and the specific maintenance standards of various airlines. The aircraft maintenance considered in AMRP is mainly daily checks, which are usually performed at night ([Liang et al., 2015](#)). Importantly, only certain [airports](#) designated as maintenance stations possess the capability to perform these maintenance tasks. The goal of aircraft maintenance routing is generally to minimize the total cost without violating the statutory remaining flight hours of each aircraft and under the resource constraints (available man hours and maintenance periods) of the maintenance station. [Wen et al. \(2022\)](#) studied the aircraft maintenance routing and flight scheduling problem at the daily operational level. The authors introduced a flexible maintenance recourse strategy to mitigate the negative impacts of operational disruptions on original aircraft maintenance routing. The available maintenance man-hours are changeable with additional expenditure. A column generation-based algorithm was developed to solve the model with high-quality solutions. Furthermore, due to its efficiency, the column generation algorithm has been widely adopted by researchers in various studies ([He et al., 2024](#), [Gao et al., 2023](#), [Jiang and Yang, 2023](#)).

In the context of AMRP, the maintenance capacity constraints of service stations are critical ([Liang et al., 2018](#); [van Kessel et al., 2023](#); [He et al., 2023](#)). Without accounting for these limitations, the AMRP may schedule maintenance at airports lacking sufficient capacity, leading to waiting time for maintenance and further flight delays. However, Due to the complexity of AMRP, most of the existing AMRP studies incorporate maintenance resource constraints into the model and assume the capacity of maintenance resources is fully ready before the start of the planning horizon. [Ruan et al. \(2021\)](#) developed a reinforcement learning-based algorithm to solve the AMRP with the consideration of a given number of maintenance workforces at each maintenance station. [Yurek \(2024\)](#) studied the AMRP at the operational level where both high-time aircraft and nonhigh-time aircraft are involved. The maintenance resources are assumed to be always available at the maintenance airports. A combinatorial benders decomposition approach was developed to solve the problem. Some scholars have explored the possibility of outsourcing part or all maintenance tasks to professional aircraft MSPs in their AMRP models. For instance, [Qin and Ng \(2023\)](#) studied the impact of cooperation between the airline and the MSP at the operational level. The authors considered a more realistic scenario under the MRO outsourcing mode which the maintenance resources are not always enough to meet all the maintenance service demands in time. They mimic the negotiation process between the two entities through an information-sharing mechanism. The values of the information sharing in terms of demand urgency level were evaluated through simulation. Different from existing literature, our AMRP model integrates maintenance technician and aircraft maintenance hangar space constraints at maintenance stations under the MRO mode, and further delves into the resource allocation strategies of MSPs when the maintenance resources are too scarce to meet all the maintenance services at the same period. In addition, we consider scenarios where, in the cases of a shortage of standard maintenance resources, airlines can access flexible maintenance resources from the MSP at an additional cost.



2.2. Resource allocation of maintenance service providers

On average, 9%–10% of the total cost of airlines goes to aircraft maintenance, which is equivalent to about \$2.5 million per aircraft per year (Deng et al., 2021). According to Prentice et al. (2022), the total market MRO market size was estimated to grow annually by 2.6% from its pre-pandemic levels in 2019, reaching \$126.9 billion by 2032. Outsourcing airline MRO operations to independent maintenance service companies has become an economical and attractive operational model (Qin and Ng, 2023). Outsourcing aircraft maintenance allows airlines to focus more on enhancing their competitiveness (Karar et al., 2023). According to the estimation by Marcontell (2013), the percentage of outsourcing of MRO activities increased by around 45% from 25% between the mid-1990s and 2012. Nam et al. (2023) emphasized the importance of focusing on technicians in outsourced maintenance operations to ensure aviation safety, while Shaukat et al. (2020) also highlighted the impact of aircraft maintenance hangar space on the feasibility of aircraft maintenance schedules. Hence, in planning aircraft routing, airlines must take into account the resource limitations imposed by MSPs. Models that focus on resources should also account for the variability inherent in these resources.

For independent MRO service providers, the task of efficiently fulfilling the growing maintenance demands of customers is a formidable challenge, primarily due to constraints in resource availability. Chen et al. (2017) emphasized that any maintenance plan must hinge on the license of technicians. They focused on optimizing the deployment of licensed technicians within an aircraft maintenance hangar operated by a solitary airline, assuming a fixed hangar capacity. Niu et al. (2022) discussed the scheduling issues of maintenance technicians from the perspective of maintenance outsourcing and constructed a technician scheduling model that considers maintenance technician costs and job satisfaction. Tseremoglou and Santos (2024) pointed out that maintenance optimization is a challenging problem facing multiple constraints such as technicians, aircraft maintenance hangar space, and availability. Besides technicians, the hangar capacity is a crucial resource for independent aircraft maintenance firms. From this perspective, Qin et al. (2018) investigated the limitations imposed by hangar capacity. They developed a model specifically designed for MSPs, aimed at optimizing the allocation of aircraft stands. Our study simultaneously considers the constraints of both maintenance technician resources and aircraft maintenance hangar space resources of MSPs in optimizing the allocation of maintenance resources.

The MSPs typically offer their aircraft maintenance services to a variety of airlines, some of which may compete with each other. Aircraft routing planning is intricately linked with the resource allocation strategies of independent aircraft MSPs. This connection arises from the intertwined needs and demands of both airlines and MSPs (Eltoukhy et al., 2022). If an airline's aircraft fails to reach the assigned maintenance station within the scheduled timeframe, it could lead to underutilization of maintenance resources. Conversely, if the independent MSP does not allocate sufficient resources for the maintenance task, this shortfall could potentially result in flight delays. Nevertheless, to our knowledge, few studies simultaneously consider the interdependence between these two key stakeholders. In our study, we simultaneously consider the allocation of technicians' resources and aircraft maintenance hangar spaces of MSPs.

2.3. Bilevel programming

Dinis et al. (2019) pointed out that MRO companies and airlines have different interests and objectives. Airlines primarily focus on maintaining or restoring the inherent reliability of aircraft at the lowest cost to ensure their airworthiness, thus guaranteeing operational efficiency and safety. In contrast, MRO companies concentrate on maximizing profits and achieving high levels of service quality through efficient services and cost control. Hierarchical optimization problems involving two decision-makers are often modeled using bilevel programming (Medina-González et al., 2021). It is particularly well-suited for real-world applications, owing to its strong modeling capabilities (Kleinert et al., 2021). Allahviranloo et al. (2019) applied bilevel programming to model a crowd-oriented autonomous driving service, where users can purchase future time slots to guarantee service. They developed a bilevel model with fleet size and vehicle routing optimization at the upper level and a set of VRP variants for household activity scheduling at the lower level. Peng et al. (2024) proposed a collaborative crowdsourcing model based on on-demand mobility (COM) services, utilizing the excess capacity of taxis and ride-hailing vehicles to transport O2O parcels, thereby alleviating the delivery pressure of parcel delivery service providers (PSP) and enhancing the profitability of on-demand mobility service provider (OMPs). By constructing a bilevel optimization model, they studied the outsourcing pricing game between PSP and OMP within the collaborative COM framework.

Although a bilevel program is capable of capturing the interactive dynamics between two different decision-makers, it can be quite intricate to solve the model to optimality. Bilevel programming remains an NP-hard problem, even when both levels are linear (Zhang et al., 2023). A prevalent approach to address this challenge is using KKT (Karush-Kuhn-Tucker) conditions to reformulate the bilevel problem into a single-level problem (Fontaine and Minner, 2014, Wang et al., 2022, Serrano et al., 2024). However, the application of KKT conditions is limited to lower-level problems that are both convex and regular. In this context, Fontaine & Minner (2014) used Benders decomposition on bilevel problems with upper-level mixed integers and a lower-level continuous linear program, splitting the problem into master and subproblems. By fixing integers, subproblems became bilevel

linear programs, solved with KKT-based reductions. Zhao et al. (2022) created a bilevel programming model for optimizing electric vehicle charging networks and logistics planning. The model features integer and continuous variables at upper and lower levels, respectively. By fixing lower-level integers and applying KKT conditions, they simplified the bilevel problem into a single-level linear programming model. Similarly, Li et al. (2023) proposed a bilevel model to capture the dynamic decision-making process between biofuel producers (acting as Stackelberg leaders) and farmers (acting as Stackelberg followers). This model considers both discrete and continuous decisions from both sides. By enumerating the discrete decisions of the follower and reformulating the optimality conditions, the original bilevel mixed-integer programming (MIP) problem is effectively converted into a solvable single-level model. However, our model exhibits significant differences from the aforementioned study. Specifically, in our model, all decision variables at both the upper and lower levels are either integer or binary, or state variables derived from binary decisions. Consequently, the approach of transforming the bilevel MIP into a single-level model is not applicable in our context.

In the aforementioned studies, the original bilevel problems were reformulated into single-level problems for solving. This process not only involves the transformation of the problem's structure but also entails imposing a series of strict restrictions on the structure of the problem. The solution techniques for continuous bilevel programming are no longer applicable to discrete bilevel programming (Mintz et al., 2023). However, an early study can be found in Bard and Moore (1992), where both upper- and lower-level variables are restricted to be binary. They proposed an algorithm that implicitly enumerates upper-level variables and solves the associated lower-level problems to obtain bilevel feasible solutions. Forghani et al. (2020) provided a bilevel mixed integer programming model for the partial interdiction of nested capacitated hierarchical facilities. To solve this model, they proposed an enumeration algorithm and a heuristic process, wherein they generated all interdiction patterns and used CPLEX to solve the lower-level problem for each pattern. Furthermore, Parajuli et al. (2021) applied implicit enumeration techniques to binary search trees to solve a three-level mixed integer programming model. This approach helped to analyze the impact of congestion on protection decisions in supply networks, aiming to mitigate major disruptions. Liu et al. (2021) proposed an enhanced branch and bound algorithm for the solution of bilevel integer programming problems where both upper-level and lower-level variables are bounded. However, they emphasized that with non-unique lower-level optima, their method might yield suboptimal results. This is because improved branching could exclude potential bilevel optimal solutions. Differently, our study establishes a novel discrete bilevel program to model the interactive dynamics between the resource allocation of an MSP and the aircraft maintenance routing of an airline company. Besides, a mutual benefit cooperative mechanism is incorporated into the bilevel optimization framework. A novel adaptive global search algorithm integrated with a column generation framework is proposed to solve the model, which is verified through the extensive computational experiments that Pareto improvement solutions are obtained for both parties.

2.4. Research gap and contribution

In most of these extant studies, the airline is typically designated as the upper-level decision maker. It is generally assumed that the airline's urgent maintenance requirements require the MSP to deploy appropriate technicians and allocate an aircraft hangar space. Nevertheless, a significant shortcoming of such model setups is the frequent neglect of the critical real-world constraint of limited maintenance resources.

Against this backdrop, this study makes a significant contribution by proposing a novel interactive decision-making framework. In this framework, considering the MSP's limited maintenance resources that impede its ability to promptly satisfy all demands from diverse airlines, we set the MSP to be the role of the strategic "leader", with the airline acting as the "follower". The proposed bilevel Mixed-Integer Programming (MIP) model is meticulously designed to capture the interactive decision-making dynamics between the MSP and the airline in real-world scenarios. In addition, the cooperative strategy embedded within this bilevel MIP model empowers the decision-making system to achieve Pareto improvements, thereby enhancing the overall efficiency and effectiveness of the decision-making process between the MSP and the airline.

3. Problem statement and formulation

In aviation operations, the MSP, as an upstream player, undertakes critical maintenance tasks for multiple airlines simultaneously. As the downstream airline, its aircraft must obey the restrictions stipulated by the civil aviation administration in different countries. At the operational level, the MSP interacts with the airlines closely to make sure the aircraft can be operated efficiently and safely. The MSP provides spatiotemporal maintenance resource information with specific prices to his clients (i.e., multiple airlines), and the airlines react accordingly based on the resource information and its prices to minimize its operational costs. Therefore, we consider a bilevel programming model to characterize the endogenous interactive decision-making structure between the MSP and his client (i.e. the airline). Section 3.1 formulates the upper-level model for the upstream MSP, Section 3.2 provides the detailed formulation for the downstream airline. Section 3.3 provides the complete bilevel mixed integer programming (MIP) model.



3.1. Upper-level model – the RA model

3.1.1. Problem statement of the RA model

In the upper-level model, the MSP needs to efficiently allocate the limited maintenance resources to meet the aircraft maintenance demands from the airline to maximize his total profit while considering the benefit of his customer. At the beginning of the planning horizon T , the MSP has information on the set of aircraft J at operations that needs maintenance service. Here, the maintenance resources refer to the available hangar space at each maintenance station ($\mathbf{m} \in \mathbf{M}$), the available man-hours at maintenance station m on day t ($\mathbf{t} \in \mathbf{T}$), and the flexible man-hours available at maintenance station m on day t . The MSP needs to decide the maintenance resource allocation, that is, whether to provide the maintenance service for aircraft j at maintenance station m on day t . Maintenance resource allocation decisions obey the priority restriction for regular maintenance and spatiotemporal resource constraints. Specifically, to guarantee the benefits of airlines, the priority restriction stipulates that flexible maintenance man-hours should be deployed after the regular maintenance man-hours are fully deployed (i.e. $\psi_{mt} = 1$). Spatiotemporal resource constraints guarantee that maintenance tasks can only be carried out when both the aircraft maintenance hangar space N_{mt} and maintenance man-hours Q_{mt} (Q_{mt}^a) at station m on day t simultaneously meet the minimum required maintenance man-hours q and the maintenance hangar space. To formulate the upper-level model, we make the assumptions as below:

- (1) The MSP can offer both regular and flexible maintenance services.
- (2) The flexible maintenance services necessitate additional resources, such as specialized maintenance technicians, which in turn results in extra costs for the MSP. A premium price coefficient, denoted as α , is introduced to compensate the extra cost associated with flexible maintenance services.
- (3) An opportunity cost coefficient, C_F , is defined to capture the impact of flight cancellations stemming from the unavailability of maintenance services. This coefficient is assumed to be fixed and known to both the MSP and the airline, which is deterministic and quantifiable based on historical data.
- (4) Flexible maintenance man-hours cannot be assigned unless all regular maintenance man-hours have been fully deployed.
- (5) The aircraft maintenance tools and spare parts are available in the planning horizon.

The notations and decision variables involved in the upper-level RA model are summarized in [Table 2](#).

Table 2. Notation for RA Model.

Set Definition	
$\mathbf{M}$	Set of maintenance stations, indexed by $\mathbf{m}$.
$\mathbf{J}$	Set of aircraft, indexed by $\mathbf{j}$.
$\mathbf{T}$	Set of days within the planning horizon, indexed by $\mathbf{t}$.
Parameters	
c_{jmt}	Regular maintenance price of aircraft j at maintenance station m on day t .
c_{mt}	The unit cost of maintenance man-hours at maintenance station m on day t .
c_{mt}^a	The unit cost of flexible maintenance man-hours at maintenance station m on day t .
d_{mt}	The unit cost of arranging aircraft maintenance hangar space at the maintenance station m on day t .
Q_{mt}	The number of man-hours at maintenance station m on day t .
Q_{mt}^a	The number of flexible man-hours at maintenance station m on day t .
N_{mt}	The number of aircraft maintenance hangar spaces at maintenance station m on the day t .
q	The number of man-hours required for the aircraft maintenance task.
$C_F \alpha$	Unit opportunity cost due to flight cancellations. Premium price coefficient for the flexible maintenance service.
x_{jmt}	Aircraft regular maintenance decisions made by the lower-level airline.
z_{jmt}	Aircraft flexible maintenance decisions made by the lower-level airline.



Set Definition

Set of maintenance stations, indexed by .

γ_{ft} Flight cancellation decisions coming from lower-level model.

Decision Variables

y_{jmt} Integer variable, the number of man-hours configured for aircraft j at maintenance station m on day t .

y_{jmt}^a Integer variable, the number of flexible man-hours configured for aircraft j at maintenance station m on day t .

w_{jmt} Binary variable, = 1, if the aircraft maintenance hangar space is configured for aircraft j at maintenance station m on day t ; 0, otherwise.

p_{jmt} Binary variable, = 1, if regular man-hours are deployed for aircraft j at maintenance station m on day t ; 0, otherwise.

p_{jmt}^a Binary variable, = 1, if flexible man-hours are deployed for aircraft j at maintenance station m on day t ; 0, otherwise.

Dependent Variables

ψ_{mt} Binary variable, = 1, if the regular man-hours of the maintenance station m has been fully deployed on day t ; 0, otherwise.

3.1.2. Mathematical modeling for the RA model

The following formulates the upper-level RA model for the MSP.

RA model:

$$\max_{\mathbf{P}, \mathbf{W}, \mathbf{Y}, \mathbf{Z}} F(\mathbf{p}, \mathbf{w}, \mathbf{y}, \mathbf{z}, \boldsymbol{\gamma}) = \sum_{m \in M} \quad (3.1)$$

$$\sum_{t \in T} \left(\sum_{j \in J} \left(c_{jmt} T F T_j x_{jmt} p_{jmt} + \alpha c_{jmt} T F T_j z_{jmt} p_{jmt}^a \right) - \sum_{j \in J} \left(y_{jmt} c_{mt} + y_{jmt}^a c_{mt}^a \right) - \sum_{j \in J} d_{mt} w_{jmt} \right) - C_F \sum_{t \in T} \sum_{f \in F^t} \gamma_{ft}$$

s.t.

$$\sum_{j \in J} y_{jmt} \leq Q_{mt} \quad \forall m \in M, t \in T \quad (3.2)$$

$$\sum_{j \in J} y_{jmt} \leq Q_{mt} - 1 + N \psi_{mt} \quad \forall m \in M, t \in T \quad (3.3)$$

$$\sum_{j \in J} y_{jmt} \geq Q_{mt} \psi_{mt} \quad \forall m \in M, t \in T \quad (3.4)$$

$$\sum_{j \in J} y_{jmt}^a \leq Q_{mt}^a \psi_{mt} \quad \forall m \in M, t \in T \quad (3.5)$$

$$\sum_{j \in J} w_{jmt} \leq N_{mt}, \quad \forall m \in M, t \in T \quad (3.6)$$

$$y_{jmt} = Q p_{jmt}, \quad \forall m \in M, t \in T, j \in J \quad (3.7)$$

$$y_{jmt}^a = Q p_{jmt}^a, \quad \forall m \in M, t \in T, j \in J \quad (3.8)$$

$$p_{jmt} + p_{jmt}^a = w_{jmt}, \quad \forall m \in M, t \in T, j \in J \quad (3.9)$$

$$w_{jmt}, \psi_{mt}, p_{jmt}, p_{jmt}^a \in \{0, 1\}, \quad \forall m \in M, t \in T, j \in J \quad (3.10)$$

$$y_{jmt}, y_{jmt}^a \in \mathbb{Z}^+, \quad \forall m \in M, t \in T, j \in J \quad (3.11)$$

The objective function (3.1) defines the profit of the MSP. Specifically, the first term is the total maintenance revenue, the second term includes all maintenance technician salaries (covering both regular and flexible maintenance technicians), the third term covers all costs of using aircraft maintenance hangar space. The fourth term accounts for the potential opportunity costs, which is defined based on the flight cancellation. The proposed objective function pushes the service provider to deploy its maintenance resources effectively and smartly, aiming at reducing the number of flights canceled due to the inability to conduct the maintenance task for the corresponding aircraft. Hence, the MSP needs to balance the maintenance service revenue, maintenance costs, and opportunity costs to achieve mutual benefit.

Constraints (3.2) – (3.5) specify the priority of maintenance resources: flexible maintenance man-hours are deployed only if all regular maintenance man-hours have been deployed at the specific maintenance station each day. Specifically, constraint (3.2) ensures that the number of deployable regular maintenance man-hours does not exceed its maximum available number at the specific maintenance station each day. Constraints (3.3) and (3.4) collectively determine whether regular maintenance man-hours have been fully deployed to the particular maintenance station each day. When $\psi_{mt} = 0$ indicates that regular

maintenance man-hours have not been fully deployed. The number of regular man-hours deployed should be less than or equal to $Q_{mt} - 1$. Conversely, when $\psi_{mt} = 1$, it indicates that regular maintenance man-hours are fully deployed, and the number of regular man-hours deployed equals Q_{mt} (see constraints (3.2) and (3.4)). Moreover, constraint (3.5) models the deployment mechanism for flexible maintenance man-hours, namely, when $\psi_{mt} = 1$, the deployment of flexible man-hours is restricted by the available quantity at station m on day t ; when $\psi_{mt} = 0$, the number of flexible man-hours deployed is 0, which ensures the deployment priority of regular maintenance man-hours. Constraint (3.6) provides that the aircraft maintenance hangar space resources at the maintenance station m on day t do not exceed the respective resource limits.

Aircraft maintenance requires sufficient aircraft maintenance hangar space and the corresponding maintenance man-hours. Constraints (3.7) – (3.9) ensure the spatiotemporal constraints of maintenance resources: that is, both aircraft maintenance hangar space and maintenance man-hours must be available simultaneously for the effective allocation of maintenance resources. (3.10) and (3.11) define the domains of the decision variables.

3.2. Lower-level model – the arc-based AMRP

3.2.1. Problem statement of the arc-based AMRP

As the downstream airline, given the allocation of the MSP on the spatiotemporal maintenance resources, she aims to minimize the total operations cost by optimizing aircraft maintenance routing decisions under the restrictions on aircraft maintenance requirements and flight tasks. First, suppose the airline operates a fleet of aircraft, denoted as the set J . Within an operational planning horizon $T = \{1, 2, \dots, |T|\}$ ($|T|$ refers to the maximum days within one planning horizon), the fleet undertakes a set of flight legs, denoted as F . The set of flight legs for each day is denoted as F^t where $t \in T$ represents a specific operational day. The decisions to be made include:

Aircraft Routing: Each aircraft is assigned with a set of connectable flight legs so as to serve as all the flights in the planning horizon. When an aircraft executes flight legs, they are represented by daytime and overnight flight arcs to form the aircraft's route. These two types of flight arcs adhere to the airport location and minimum sit time ST constraints. Specifically, if an aircraft sequentially executes flight legs f^t and g^r that forms the flight arc (f^t, g^r) , it must satisfy: (1) the arrival airport of the flight leg f^t must be the departure airport of its successive flight leg g^r ; (2) the arrival time of the flight leg f^t must be earlier than the departure time of its successive flight leg g^r , and the time interval between them is greater than the specified minimum sit time ST . Each flight leg can either be operated by aircraft j that incurs the operational costs c_{jf} , or be canceled with the cancellation cost c_{cf} .

Performance of Aircraft Maintenance: Each aircraft must undergo maintenance before exceeding its maximum allowed flight hours H_{max} , or number of takeoffs F_{max} . Maintenance tasks depend on the execution of flight legs. They can be carried out at least after the aircraft completes the initial flight leg during its route within an overnight flight arc and lands at an airport equipped with a maintenance station. Performing maintenance tasks depends on the availability of maintenance resources each day (i.e., maintenance man-hours and aircraft maintenance hangar space at a maintenance station). After performing the maintenance task, the aircraft's flight hours and number of takeoffs are reset to 0 (i.e., $TFT_{jt} = 0$ and $TFN_{jt} = 0$).

To formulate the lower-level model, we make the assumptions as below:

- (1) The maintenance task is carried out during nighttime.
- (2) The flight is canceled if it is not assigned to any aircraft available in the planning horizon.

The notations and decision variables involved in the arc-based AMRP model are summarized in Table 3.

Table 3. Notation for the Arc-Based AMRP Model.

Set and Indices Definition		
J	Set of aircraft, indexed by j .	
M	Set of maintenance stations, indexed by m .	
T	Set of days within the planning horizon, indexed by t, τ .	
F	Set of all flights within the planning horizon, indexed by f .	
F^t	Set of flights on day t , indexed by f^t, g^t .	
L	Set of airports, indexed by l .	

Set and Indices Definition

Parameters

K	Average time required to complete the maintenance task.
H_{max}	Maximum flight hours between two consecutive maintenance tasks.
F_{max}	Maximum number of takeoffs between two consecutive maintenance tasks.
ST	Minimum sit time between connected flights.
FT_j^I	Total flight hours of aircraft j at the beginning of the planning horizon.
FN_j^I	Number of takeoffs of aircraft j at the beginning of the planning horizon.
N	A tremendous positive value.
n_j	The number of man-hours required to maintain aircraft j .
$s_j (e_j)$	The original airport for aircraft j , represented by two symbols: s_j for departure node at the beginning of the planning horizon and e_j for the arrival node at the end of the planning horizon.
d_{f^t}	Scheduled departure time of flight f^t .
a_{f^t}	Scheduled arrival time of flight f^t .
$d_{f^t l}$	=1, if the departure airport of the flight f^t is l ; =0, otherwise.
$a_{f^t l}$	=1, if the arrival airport of the flight f^t is l ; =0, otherwise.
$\beta_{f^t m}$	=1, if the arrival airport of the flight f^t has maintenance station m ; = 0, otherwise.
c_{jmt}	Regular maintenance cost of aircraft j at maintenance station m on day t .
c_{jf^t}	The operating cost of aircraft j operating flight leg f^t .
c_{f^t}	The cancellation cost of flight leg f^t .
α	Maintenance extra payment coefficient.

Decision Variables

$v_{jf^t g^r}$	Binary variable, = 1, if aircraft j has completed flight leg f^t and then executes the flight leg g^r ; 0, otherwise.
x_{jmt}	Binary variable, = 1, if aircraft j is undergoing regular maintenance at maintenance station m on day t ; 0, otherwise.
z_{jmt}	Binary variable, = 1, if aircraft j is undergoing flexible maintenance at maintenance station m on day t ; 0, otherwise.
γ_{f^t}	Binary variable, = 1, if flight f^t is canceled; 0, otherwise.
$\omega_{jf^t g^{t+1}}$	Auxiliary binary variable, = 1, if aircraft j is maintained after flight f^t and then executes flight g^{t+1} ; 0, otherwise.

Dependent Variables

FT_{jt}	The cumulative flight hours of aircraft j on day t .
TFT_{jt}	The cumulative flight hours of aircraft j up to day t .
FN_{jt}	The cumulative number of takeoffs of aircraft j on day t .
TFN_{jt}	The cumulative number of takeoffs of aircraft j up to day t .

3.2.2. Mathematical modeling for arc-based AMRP

Given the notation, the arc-based lower-level AMRP model is formulated as follows.

Arc-based AMRP model:

$$\min_{X, \Omega, V, Z, Y} f(v, \omega, x, z, \gamma) = \sum_{j \in J} \sum_{t \in T} \sum_{r \in T} \sum_{f^t \in F^t} \sum_{g^r \in F^r} c_{jf^t} v_{jf^t g^r} + \sum_{j \in J} \sum_{m \in M} \sum_{t \in T} c_{jmt} TFT_{jt} x_{jmt} + \sum_{j \in J} \sum_{m \in M} \sum_{t \in T} \alpha c_{jmt} TFT_{jt} z_{jmt} + \sum_{t \in T} \sum_{f^t \in F^t} c_{f^t} \gamma_{f^t} \quad (3.12)$$



New

s.t.

$$\sum_{j \in J} \left(\sum_{g^t \in F^t} v_{j f^t g^t} + \sum_{g^{t+1} \in F^{t+1}} v_{j f^t g^{t+1}} \right) + \gamma_{f^t} = 1, \quad \forall t \in T \setminus \{|T|\}, f^t \in F^t \quad (3.13)$$

$$\sum_{f^1 \in F^1} v_{j s_j f^1} d_{f^1 s_j} = 1, \quad \forall j \in J \quad (3.14)$$

$$\sum_{f^{|T|} \in F^{|T|}} v_{j f^{|T|} e_j} a_{f^{|T|} e_j} = 1, \quad \forall j \in J \quad (3.15)$$

$$\sum_{f^t \in F^t \cup \{s_j\}} v_{j f^t g^t} = \sum_{f^t \in F^t \cup \{e_j\}} v_{j g^t f^t} + \sum_{f^{t+1} \in F^{t+1} \cup \{e_j\}} v_{j g^t f^{t+1}}, \quad \forall j \in J, t \in T, \tau \in \{|T|\}, f^t \in F^t, g^t \in F^t \quad (3.16)$$

$$d_{g^t} - a_{f^t} - ST \geq N(v_{j f^t g^t} - 1), \quad \forall j \in J, t \in T, f^t \in F^t, g^t \in F^t \quad (3.17)$$

$$\sum_{j \in J} v_{j f^t g^t} \leq \sum_{l \in L} a_{f^t l} d_{g^t l}, \quad \forall t, \tau \in T, f^t \in F^t, g^t \in F^t, f^t \neq g^t \quad (3.18)$$

$$FT_{jt} = \sum_{f^t \in F^t} \left(\sum_{g^t \in F^t, g^t \neq f^t} v_{j f^t g^t} (a_{f^t} - d_{f^t}) \right), \quad \forall j \in J, t \in T, \quad (3.19)$$

$$TFT_{jt} = \begin{cases} FT_{jt} + FT_j^I & \forall j \in J, t = 1 \\ TFT_{jt-1} (1 - \omega_{j f^{t-1} g^t}) + FT_{jt} & \forall j \in J, t = 2, 3, \dots, T \end{cases} \quad (3.20)$$

$$FN_{jt} = \sum_{f^t \in F^t} \sum_{g^t \in F^t, g^t \neq f^t} v_{j f^t g^t}, \quad \forall j \in J, t \in T \quad (3.21)$$

$$TFN_{jt} = \begin{cases} FN_{jt} + FN_j^I & \forall j \in J, t = 1 \\ TFN_{jt-1} (1 - \omega_{j f^{t-1} g^t}) + FN_{jt} & \forall j \in J, t = 2, 3, \dots, T \end{cases} \quad (3.22)$$

$$TFT_{jt} \leq H_{max}, \quad \forall j \in J, t \in T \quad (3.23)$$

$$TFN_{jt} \leq F_{max}, \quad \forall j \in J, t \in T \quad (3.24)$$

$$d_{g^{t+1}} \geq a_{f^t} + ST + K - N(1 - \omega_{j f^t g^{t+1}}), \quad \forall j \in J, t \in T \setminus \{|T|\}, f^t \in F^t, g^{t+1} \in F^{t+1} \quad (3.25)$$

$$\sum_{m \in M} v_{j f^t g^{t+1}} \beta_{f^t m} \geq \omega_{j f^t g^{t+1}}, \quad \forall j \in J, t \in T \setminus \{|T|\}, f^t \in F^t, g^{t+1} \in F^{t+1} \quad (3.26)$$

$$\omega_{j f^t g^{t+1}} \beta_{f^t m} = x_{jmt} + z_{jmt}, \quad \forall t \in T, j \in J, m \in M \quad (3.27)$$

$$\sum_{m \in M} \sum_{t \in T} (x_{jmt} + z_{jmt}) \leq 1, \quad \forall j \in J \quad (3.28)$$

$$\sum_{j \in J} x_{jmt} \leq \sum_{j \in J} w_{jmt}, \quad \forall t \in T, m \in M \quad (3.29)$$

$$\sum_{j \in J} x_{jmt} n_j \leq \sum_{j \in J} y_{jmt}, \quad \forall t \in T, m \in M \quad (3.30)$$

$$\sum_{j \in J} z_{jmt} n_j \leq \sum_{j \in J} y_{jmt}^c, \quad \forall t \in T, m \in M \quad (3.31)$$

$$\gamma_{f^t} \in \{0, 1\}, \quad \forall t \in T, f^t \in F^t \quad (3.32)$$

$$v_{j f^t g^t} \in \{0, 1\}, \omega_{j f^t g^t} \in \{0, 1\}, \quad \forall j \in J, t \in T, f^t \in F^t, g^t \in F^t \quad (3.33)$$

$$x_{jmt} \in \{0, 1\}, z_{jmt} \in \{0, 1\}, \quad \forall j \in J, m \in M, t \in T \quad (3.34)$$

$$FT_{jt}, TFT_{jt} \in \mathbb{R}^+, FN_{jt}, TFN_{jt} \in \mathbb{Z}^+ \quad \forall j \in J, t \in T \quad (3.35)$$

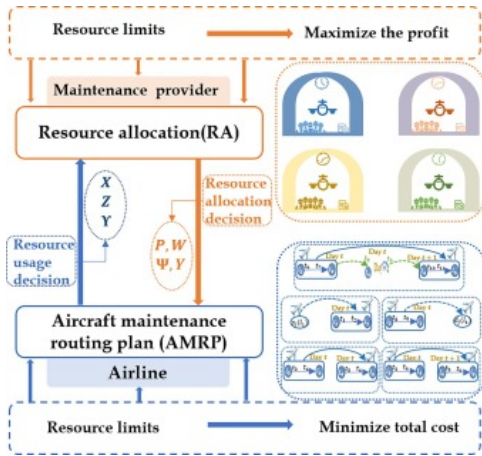
The objective function (3.12) aims to minimize the total operating costs, which include flight operating costs, regular maintenance costs, flexible maintenance costs, and flight cancellation costs. The constraint (3.13) ensures that each flight is either covered by an aircraft or canceled. Flight execution can be achieved through either a daytime flight arc or an overnight flight arc. This distinction is made in the model by $y_{j f^t g^t}$ with $\tau = t$ for a daytime flight arc and $\tau = t + 1$ for an overnight flight arc (see (3.13)). Please note that the daytime and overnight flight arcs are considered without explicit expression in subsequent constraints. Constraints (3.14) and (3.15) ensure that all aircraft must depart from and return to their original airports. Constraint (3.16) guarantees the flow balance constraint for the aircraft route. Constraint (3.17) indicates the relationship between the departure time and the arrival time for two consecutive flights, while constraint (3.18) ensures the airport location

restriction for two consecutive flights. Constraints (3.19) and (3.20) define the total flight hours on day t and the cumulative flight hours up to day t , respectively, with the latter being reset to 0 after performing the maintenance task. Similarly, constraints (3.21) and (3.22) calculate the corresponding cumulative number of takeoffs for each aircraft j .

Constraints (3.23) and (3.24) ensure that within the planning horizon, each aircraft's flight hours and the number of takeoffs obey the official safety restrictions. Constraint (3.25) pertains to the timing of aircraft maintenance. Specifically, if the maintenance task is performed during an overnight flight arc (f^t, g^{t+1}) , the departure time of g^{t+1} must be later than the arrival time of f^t plus the sit time and maintenance time. Constraint (3.26) restricts the condition that the aircraft performs its maintenance task. It stipulates that maintenance activities can only occur after an aircraft has completed the flight leg f^t of an overnight flight arc (f^t, g^{t+1}) and arrives at the airport equipped with the maintenance station. Constraint (3.27) indicates that the aircraft can have at most one kind of maintenance service option, i.e., regular, or flexible maintenance service at maintenance station m on day t . Constraint (3.28) ensures at most one maintenance per aircraft within the planning horizon. Constraints (3.29), (3.30), and (3.31) respectively impose limits on maintenance resources in terms of the number of aircraft maintenance hangar spaces at the maintenance station, regular maintenance man-hours, and the flexible maintenance man-hours for each maintenance station on each day. Constraints (3.32)–(3.35) define the domains for each decision variable.

3.3. Bilevel model

Based on the MSP's RA model (Section 3.1) and the airline's AMRP model (Section 3.2), we propose a novel bilevel mixed-integer programming model formulated as below. To differentiate upper-level and lower-level decisions and reflect their endogenous interactions, the upper-level variables are highlighted in orange, while the lower-level variables are highlighted in blue. Fig. 2 illustrates the endogenous interactions between the MSP and the airline at the operational level.



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Fig. 2. Schematic Diagram of the Proposed Bilevel Mixed-integer Programming.

Bilevel MIP model:

$$\max F(P, W, \Psi, Y, X, Z, Y)$$

$$\text{s.t. } G(P, W, \Psi, Y) \leq 0, P, W, \Psi \in \{0, 1\}^{|J| \times |M| \times |T|}, Y \in \mathbb{Z}_+^{|J| \times |M| \times |T|}.$$

$$(X, Z, Y) \in \mathcal{F}(W, Y)$$

$$= \argmin \left\{ f(V, \Omega, X, Z, Y): g(W, Y, V, \Omega, X, Z, Y, FT, TFT, FN, TFN) \leq 0, V, \Omega \right.$$

$$\left. \in \{0, 1\}^{|F| \times |F-1| \times |J|}, X, Z \in \{0, 1\}^{|J| \times |M| \times |T|}, FT, TFT \in \mathbb{R}_+^{|J| \times |T|}, FN, TFN \in \mathbb{Z}_+^{|J| \times |T|} \right\}$$

4. Solution approach for the bilevel model

In this paper, we propose a novel bilevel MIP model, in which the upper-level model is an IP model for the MSP, and the lower-level model is an MIP model for the airline company. Although it is widely applied in different areas, bilevel optimization is still challenging due to their inherent complexity in terms of the interactive decision making structure. In particular, there is an efficient and universal solution method and algorithm framework to solve discrete bilevel models. The optimality conditions heavily depend on the specific structure of the problem involved. In existing literature, the main approach to reformulating the bilevel problem into a single-level problem is based on high point relaxation (HPR) (Moon et al., 2022, Soares et al., 2021,

Tahernejad et al., 2020). The main idea is to combine the lower-level constraints into the upper-level model. It has been identified that HPR can only provide a weak upper bound for the upper-level model. Different specific techniques, such as branch and bound, are developed to realize convergence. Another main approach is based on an implicit enumeration scheme (Liu et al., 2021, Zeng and An, 2014, Bard and Moore, 1992).

The basic logic behind it is to convert the lower-level model into a linear programming model by implicitly enumerating the finite integer variables in the lower-level problem. In this way, the optimality conditions of the lower-level problem can be found according to the strong duality. Thus, under the enumeration scheme, the bilevel model is reformulated into a single-level model by adding the optimality conditions of the linearized lower-level problem into the upper-level problem. The premise of applying such reformulation and decomposition method is the bilevel problem demonstrates a dual block angular structure (Zeng and An, 2014). Besides, as we can see from the literature, another difficulty is to tackle the discrete variable in the lower-level problem. Once the lower-level problem can be converted into the upper-level model, it is feasible to use the existing methods to solve single-level MIP.

4.1. Greedy search nested column generation algorithm (GSNCG)

Due to the complexity of the problem structure, reformulating the bilevel model into a single-level still makes our problem intractable. The lower-level AMRP problem we explore is already an NP-hard problem (Zhang et al., 2024). In addition, the objective function of the upper-level problem cannot be blocked from the lower-level variables. Therefore, according to the features of the proposed bilevel model, we propose a novel nested algorithm to obtain an optimal bilevel solution based on the convergence mechanism proposed by Zeng and An (2014). The pseudo-code of the proposed GSNCG algorithm is shown below.

First, we solve the relaxed upper-level model, in which additional constraints are added along with (3.2)–(3.11) to relax the impacts from the lower-level problem. We define this relaxed version as a self-centered model for the MSP. The first constraint (A1-1) is a parameterized constraint converted from the upper-level objective function. The right-hand-side value equals $UB - \epsilon$. The purpose of having this parameterized constraint is to realize the convergence of the UB. ϵ is a small tolerance that is added to the right-hand side of the constraint to overcome computational difficulties due to strict inequality. The additional constraints (A1-2)–(A1-5) relax the feedback from the lower-level model. In this way, the MSP will try his best to allocate as many maintenance resources to the airline company to earn more revenue. In this relaxed scenario, the MSP focuses on its benefit without considering the benefit of the airline company. Constraint (A1-4) aims to ignore the opportunity cost in the relaxed upper-level model. Constraint (A1-5) guarantees the profitability of any maintenance resource allocation decision made by the MSP. After solving the relaxed upper-level RA model Π^* , we obtain an upper bound UB for the profit of the MSP and the corresponding solution $(P^{I^*}, W^{I^*}, Y^{I^*}, \Psi^{I^*}, X^{I^*}, Z^{I^*}, Y^{I^*})$. Note that, here, $(X^{I^*}, Z^{I^*}, Y^{I^*})$ are determined by Π^* through constraints (A1-2)–(A1-5). They are not the feedback from the lower-level model, but the relaxed solution corresponding to Π^* . They are even not feasible. Second, we check the current UB and LB, if $UB - LB < \epsilon + C_f^T Y^{I-1^*}$, the algorithm is terminated and we find the optimal solution. Here, $C_f^T Y^{I-1^*}$ refers to the opportunity cost determined by the lower-level optimal solution at $I-1$ iteration. Otherwise, go to the next step. Given the solution (W^{I^*}, Y^{I^*}) , we solve the lower-level model to optimality and obtain its optimal solution as $\pi(W^{I^*}, Y^{I^*})$ which is the rational reaction of the airline. Next, if we have multiple optimal solutions from the lower level, we compute the optimistic rational reaction that minimizes the upper-level objective and update $(X^{I^*}, Z^{I^*}, Y^{I^*})$ and the corresponding optimal value of the upper-level model $\Pi_0(W^{I^*}, Y^{I^*})$. Update the $LB = \max\{LB, \Pi_0(W^{I^*}, Y^{I^*})\}$. The loop process keeps proceeding until meets the stop criteria $UB - LB < \epsilon + C_f^T Y^{I-1^*}$ or reaches the maximum iteration. The relaxed upper-level problem Π^* can be directly solved by the off-the-shelf solver. In each iteration, to solve the lower-level problem to optimality in large-scale cases, a column generation-based algorithm is further developed in Section 5.

Algorithm 1: the greedy search nested column generation algorithm.

Input: model parameters.

Output: optimal solution for both MSP and airline.

– Initialization:

Set $LB = -\infty$, $UB = +\infty$, and $I = 0$.

– Loop Process:

1: while $I < I_{max}$ do

2: $I = I + 1$;

3: Solve the following relaxed upper-level RA model: $\Pi^* = \max F(P, W, Y, \Psi, X, Z, Y)$.



New

Algorithm 1: the greedy search nested column generation algorithm.

s. t. (3.2) – (3.11)

$$F(\mathbf{P}, \mathbf{W}, \mathbf{Y}, \Psi, \mathbf{X}, \mathbf{Z}, \mathbf{Y}) \leq UB - \varepsilon \quad (A1-1)$$

$$x_{jmt} \geq p_{jmt}, \forall j \in J, m \in M, t \in T \quad (A1-2)$$

$$z_{jmt} \geq p_{jmt}^a, \forall j \in J, m \in M, t \in T \quad (A1-3)$$

$$\gamma_{f^t} = 0, \forall t \in T, f^t \in F^t \quad (A1-4)$$

$$TFT_j > \frac{q_{cmt}}{c_{jmt}}, TFT_j > \frac{q_{cmt}^a}{\alpha c_{jmt}}, \forall j \in J, m \in M, t \in T \quad (A1-5)$$

4: Derive an optimal solution $(\mathbf{P}^{I*}, \mathbf{W}^{I*}, \mathbf{Y}^{I*}, \Psi^{I*}, \mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*})$, and update $UB = \Pi^*$.

5: if $UB - LB < \epsilon + \mathbf{C}_f^T \mathbf{Y}^{I-1*}$ then

6: break

7: else:

8: Solve the lower-level AMRP model by the proposed CG based algorithm for given

$(\mathbf{W}^{I*}, \mathbf{Y}^{I*})$, Let the optimal value of the lower-level problem as:

$$\begin{aligned} \pi(\mathbf{W}^{I*}, \mathbf{Y}^{I*}) = \\ \min \left\{ f(\mathbf{V}, \Omega, \mathbf{X}, \mathbf{Z}, \mathbf{Y}): g(\mathbf{W}^{I*}, \mathbf{Y}^{I*}, \mathbf{V}, \Omega, \mathbf{X}, \mathbf{Z}, \mathbf{Y}, \mathbf{FT}, \mathbf{TFT}, \mathbf{FN}, \mathbf{TFN}) \leq 0, \mathbf{V}, \Omega \right. \\ \left. \in \{0, 1\}^{|F| \times |F-1| \times |J|}, \mathbf{X}, \mathbf{Z} \in \{0, 1\}^{|J| \times |M| \times |T|}, \mathbf{FT}, \mathbf{TFT} \in \mathbb{R}_+^{|J| \times |T|}, \mathbf{FN}, \mathbf{TFN} \in \mathbb{Z}_+^{|J| \times |T|} \right\} \end{aligned}$$

9: Compute the optimistic reaction for the upper-level RA model:

$$\begin{aligned} \Pi_0(\mathbf{W}^{I*}, \mathbf{Y}^{I*}) = \\ \max \left\{ F(\mathbf{P}, \mathbf{W}, \mathbf{Y}, \Psi, \mathbf{X}, \mathbf{Z}, \mathbf{Y}) \mid f(\mathbf{V}, \Omega, \mathbf{X}, \mathbf{Z}, \mathbf{Y}) \leq \pi(\mathbf{W}^{I*}, \mathbf{Y}^{I*}), \right. \\ \left. g(\mathbf{W}, \mathbf{Y}, \mathbf{V}, \Omega, \mathbf{X}, \mathbf{Z}, \mathbf{Y}, \mathbf{FT}, \mathbf{TFT}, \mathbf{FN}, \mathbf{TFN}) \leq 0, \mathbf{V}, \Omega \in \{0, 1\}^{|F| \times |F-1| \times |J|}, \mathbf{X}, \mathbf{Z} \right. \\ \left. \in \{0, 1\}^{|J| \times |M| \times |T|}, \mathbf{FT}, \mathbf{TFT} \in \mathbb{R}_+^{|J| \times |T|}, \mathbf{FN}, \mathbf{TFN} \in \mathbb{Z}_+^{|J| \times |T|} \right\} \end{aligned}$$

10: Derive the optimal reaction $(\mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*})$, update $LB = \max\{LB, \Pi_0(\mathbf{W}^{I*}, \mathbf{Y}^{I*})\}$.

11: Update $(\mathbf{W}^{I*}, \mathbf{Y}^{I*})$.

12: end if

13: end while

14: return $LB, (\mathbf{P}^{I*}, \mathbf{W}^{I*}, \mathbf{Y}^{I*}, \Psi^{I*}), \pi(\mathbf{W}^{I*}, \mathbf{Y}^{I*}), (\mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*})$.

4.2. A Discussion on convergence

For the bilevel model, to solve it to optimality implies to solve the following optimal problem:

$$\begin{aligned} \max(F(\mathbf{P}, \mathbf{W}, \mathbf{Y}, \Psi, \mathbf{X}, \mathbf{Z}, \mathbf{Y}): (\mathbf{P}, \mathbf{W}, \mathbf{Y}, \Psi, \mathbf{X}, \mathbf{Z}, \mathbf{Y}) \in IR) \\ IR = \left\{ (\mathbf{P}, \mathbf{W}, \mathbf{Y}, \Psi, \mathbf{X}, \mathbf{Z}, \mathbf{Y}): \mathbf{P}, \mathbf{W}, \Psi \in \{0, 1\}^{|J| \times |M| \times |T|}, \mathbf{Y} \in \mathbb{Z}_+^{|J| \times |M| \times |T|}, \right. \\ \left. \{\mathbf{X}, \mathbf{Z}, \mathbf{Y}\} \in \pi(\mathbf{W}, \mathbf{Y}) \right\} \end{aligned}$$

In other words, given the iteration I , if $(\mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*})$ is the optimal solution of the lower-level problem given the optimal solution of the relaxed upper-level problem $(\mathbf{W}^{I*}, \mathbf{Y}^{I*})$, and $F^I(\mathbf{P}^{I*}, \mathbf{W}^{I*}, \mathbf{Y}^{I*}, \Psi^{I*}, \mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*}) =$, then

$$F^{I+1}(\mathbf{P}^{I+1*}, \mathbf{W}^{I+1*}, \mathbf{Y}^{I+1*}, \Psi^{I+1*}, \mathbf{X}^{I+1*}, \mathbf{Z}^{I+1*}, \mathbf{Y}^{I+1*})$$

the upper-level problem gets optimality. Thus, we have the global optimal solution.

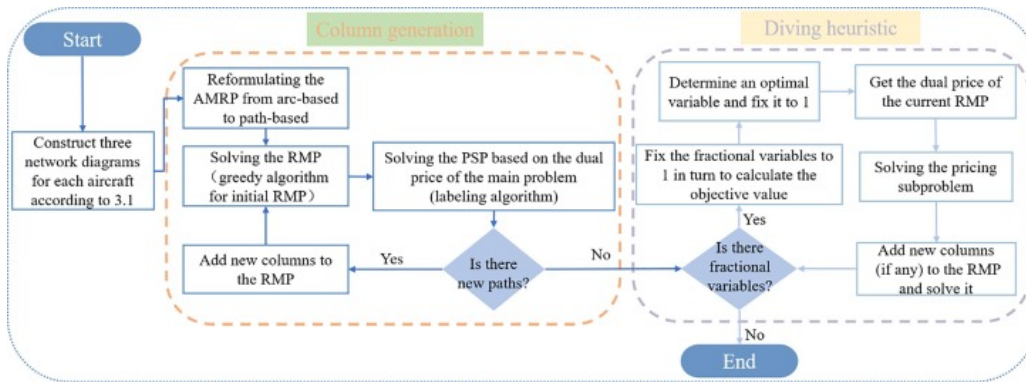
As we analyzed in Section 4.1, the relaxed upper-level model aims to allocate as much as cost-effective maintenance resources to the airline without consideration of the needs and benefits of the airline. For the iteration I , it obtains the current highest profit $\Pi^I(\mathbf{P}^{I*}, \mathbf{W}^{I*}, \mathbf{Y}^{I*}, \Psi^{I*}, \mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*}) = UB^I$. Corresponding to $(\mathbf{W}^{I*}, \mathbf{Y}^{I*})$, we compute the rational reaction, then update $(\mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*})$, $(\mathbf{W}^{I*}, \mathbf{Y}^{I*})$ as well as the current $LB^I = F^I(\mathbf{P}^{I*}, \mathbf{W}^{I*}, \mathbf{Y}^{I*}, \Psi^{I*}, \mathbf{X}^{I*}, \mathbf{Z}^{I*}, \mathbf{Y}^{I*})$. Note that there always exist feasible solutions for the lower-level problem given any solutions $(\mathbf{W}^{I*}, \mathbf{Y}^{I*})$ from the upper-level problem. This is because, for the lower-level problem, the airline can choose to cancel flights to postpone the maintenance tasks and serve the remaining

flights. In this case, a higher total operational cost is involved. Due to such good characteristics, we can compute the rational reaction as long as there is a feasible solution from the upper-level. This result guarantees that we can always find the LB for each iteration. In addition, given the iteration I , $UB^I > LB^I$, as $UB^I \geq LB^I + C_f^T Y^{I*}$. Thus, we have $UB^I > UB^{I+1} > LB^{I+1} \geq LB^I$. At $I+1$ iteration, if $(W^{I+1*}, Y^{I+1*}) = (W^{I*}, Y^{I*})$, then $F^{I+1}(P^{I+1*}, W^{I+1*}, Y^{I+1*}, \Psi^{I+1*}, X^{I+1*}, Z^{I+1*}, Y^{I+1*}) =$, namely $UB^{I+1} + C_f^T Y^{I*} = LB^I$. We meet the bilevel optimal condition, the algorithm is converged and terminated. If $(W^{I+1*}, Y^{I+1*}) \neq (W^{I*}, Y^{I*})$, the inequality $UB^{I+1} > LB^I + C_f^T Y^{I*}$ holds, the algorithm proceeds with the iteration until it meets the stop criteria. In the proposed bi-level MIP model, as both the upper and lower decision variables are integers, the total iteration of the proposed algorithm is finite. We set a maximum iteration number in case there are time limits on the computation.

5. Column generation based algorithm for AMRP

In this section, we propose a column generation framework that is designed to solve the proposed AMRP model. Within this framework, the original arc-based AMRP model is reformulated into a path-based model in the Restricted Master Problem (RMP) to improve the computational efficiency of the complex combinatorial problem. The specific subproblems are developed to generate potential aircraft paths with the consideration of arc connectivity constraints.

As shown in Fig. 3, the AMRP column generation framework initially constructs three network diagrams based on different types of connecting arcs. Subsequently, the greedy algorithm is employed to generate an initial route for each aircraft. This step is not only fundamental to solving the initial relaxed master problem but also critical in calculating the dual prices related to various constraints. Furthermore, the new paths that can help reduce the total operating costs are generated by solving the subproblems. The RMP is then resolved with additional added variables and checked with the reduced cost. The process keeps iteration until no path that can contribute to a lower objective value. Each network diagram corresponds to different maintenance scenarios: without maintenance, regular maintenance, and urgent maintenance (which corresponds to flexible maintenance service). Accordingly, we subdivided the aircraft path subproblem into three distinct subproblems that generate aircraft paths regarding different maintenance scenarios.



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Fig. 3. Solution Algorithm Flowchart for AMRP.

Once the optimal linear relaxation solution is found, we employ the diving heuristic algorithm to perform branching operations. This algorithm attempts to fix fractional variables to 1 one by one and calculates the objective value to identify the variables that can minimize the objective value. Subsequently, we recalculate the dual prices of the RMP after fixing the variables and further solving the path generation problem (i.e., subproblem) using the labeling algorithm. If the subproblem identifies new paths, these paths are added to the RMP for further solution updating. It keeps iterating until all variables are integers.

5.1 AMRP reformulation – restricted master problem (RMP), 5.2 Sub-Problem discuss the detailed reformulation processes with the restricted master problem and the subproblem, respectively. For the details on the proposed labeling algorithm to solve the subproblem, readers can refer to [Appendix B](#).

5.1. AMRP reformulation – restricted master problem (RMP)

We adhere to the symbol usage conventions established in [Section 3](#) to ensure clarity in our exposition. The notation utilized in this section is displayed in [Appendix A](#).

RMP

$$\min \left\{ \sum_{j \in J} \sum_{i \in I} \sum_{r_{ji} \in R_{ji}} c_{r_{ji}} x_{r_{ji}} + \sum_{f \in F} c_f \gamma_f \right\} \quad (5.1)$$

s.t.

$$\sum_{j \in J} \sum_{i \in I} \sum_{r_{ji} \in R_{ji}} a_{r_{ji}}^f x_{r_{ji}} + \gamma_f = 1, \quad \forall f \in F, \pi_f \quad (5.2)$$

$$\sum_{i \in I} \sum_{r_{ji} \in R_{ji}} x_{r_{ji}} \leq 1, \quad \forall j \in J, \vartheta_j \quad (5.3)$$

$$\sum_{j \in J} \sum_{i \in I} \sum_{r_{ji} \in R_{ji}} s_{r_{ji}}^{mt} x_{r_{ji}} \leq \sum_{j \in J} w_{jmt} \quad \forall m \in M, t \in T, i = 2, 3, \delta_{mt} \quad (5.4)$$

$$\sum_{j \in J} \sum_{r_{ji} \in R_{ji}} s_{r_{ji}}^{mt} n_j x_{r_{ji}} \leq \sum_{j \in J} y_{jmt} \quad \forall m \in M, t \in T, i = 2, \varphi_{mt} \quad (5.5)$$

$$\sum_{j \in J} \sum_{r_{ji} \in R_{ji}} s_{r_{ji}}^{mt} n_j x_{r_{ji}} \leq \sum_{j \in J} y_{jmt}^u \quad \forall m \in M, t \in T, i = 3, \tau_{mt} \quad (5.6)$$

$$x_{r_{ji}}, \gamma_f \in \{0, 1\} \quad \forall i \in I, j \in J, r_{ji} \in R_{ji}, \forall f \in F \quad (5.7)$$

Objective function (5.1) aims to minimize the total operational costs of the airline. Here, the cost denoted by $c_{r_{ji}}$ includes the aircraft's flight operational costs and the corresponding maintenance costs (if there is maintenance service occurred). Constraint (5.2) ensures that all flights are either operated or canceled. Constraint (5.3) guarantees that each aircraft can perform at most one path. Constraints (5.4) – (5.6) restrict maintenance resources to ensure they do not exceed available resources. Specifically, constraint (5.4) limits the aircraft maintenance hangar space in each maintenance station on a given day. In contrast, constraints (5.5) and (5.6) restrict the resources of regular maintenance and flexible maintenance man-hours, respectively. Finally, constraint (5.7) defines the feasible domain of the decision variables.

5.2. Sub-Problem

The RMP model transmits its dual prices to the subproblem, while the subproblem transfers the new aircraft routes (columns) back to the RMP model. Following the resolution of each RMP, the subproblem acquires dual price information corresponding to each constraint. Based on the dual price information, the subproblems generate the columns (i.e., aircraft paths in AMRP) with negative reduced cost for each aircraft. In the column generation phase, handling columns with negatively reduced costs is technically called the “pricing process,” which involves solving the subproblem to identify columns not yet considered in the RMP that can improve the current solution with their negatively reduced cost. In the paper, we categorize the paths into three types. The reduced cost of the r_{ji} -th route in i -th type generated for aircraft j is defined as $\theta_{r_{ji}}$. The formulas for calculating the reduced costs for different types of paths are outlined in equations (5.8)– (5.10).

$$\theta_{r_{j1}} = c_{r_{j1}} - \sum_{f \in F} a_{r_{j1}}^f \pi_f - \vartheta_j \quad (5.8)$$

$$\theta_{r_{j2}} = c_{r_{j2}} - \sum_{f \in F} a_{r_{j2}}^f \pi_f - \sum_{m \in M} \sum_{t \in T} s_{r_{j2}}^{mt} \delta_{mt} - \sum_{m \in M} \sum_{t \in T} s_{r_{j2}}^{mt} n_j \varphi_{mt} - \vartheta_j \quad (5.9)$$

$$\theta_{r_{j3}} = c_{r_{j3}} - \sum_{f \in F} a_{r_{j3}}^f \pi_f - \sum_{m \in M} \sum_{t \in T} s_{r_{j3}}^{mt} \delta_{mt} - \sum_{m \in M} \sum_{t \in T} s_{r_{j3}}^{mt} n_j \tau_{mt} - \vartheta_j \quad (5.10)$$

Specifically, the dual variables π_f , ϑ_j , δ_{mt} , φ_{mt} and τ_{mt} correspond to the constraints (5.2) – (5.6) of the RMP. The terms $c_{r_{ji}}$ represent the flight and maintenance cost (if there is maintenance service performed), respectively, for aircraft j 's execution path r_{ji} . The expression $\sum_{f \in F} a_{r_{ji}}^f \pi_f$ calculates the aggregated dual costs associated with each flight on the path. Meanwhile, the dual costs $\sum_{m \in M} \sum_{t \in T} s_{r_{ji}}^{mt} \delta_{mt}$ are related to aircraft maintenance hangar space resources at each maintenance station on the path. Further, $\sum_{m \in M} \sum_{t \in T} s_{r_{ji}}^{mt} n_j \varphi_{mt}$ and $\sum_{m \in M} \sum_{t \in T} s_{r_{ji}}^{mt} n_j \tau_{mt}$ aggregate the dual costs for regular maintenance technician resources and flexible maintenance technician resources on the path. The sub-problem in our study can be seen as a shortest path problem with resource constraints (SPPRC), which is solved by the proposed labeling setting algorithm. For the details on the subproblem-solving algorithm and the branching process, readers can refer to [Appendix A Supplementary material, Pan et al., 2023, Sadykov et al., 2019, Xiao et al., 2022](#).

6. Experimental study

The experiments were carried out on a computer with an Intel Core i7-8565U processor (1.80GHz), and 8GB of RAM, running the Windows 10 operating system. Our models were coded using Python 3.9 within the PyCharm integrated development environment. All LP and MIP problems were solved with GUROBI optimizer version 9.5.2.

In the following subsection, we first describe the settings of the test instances used in the experiment study in [Section 6.1](#). By providing this comprehensive overview of our [experimental design](#), we aim to enhance the transparency and methodological rigor of our approach. This detailed exposition not only facilitates a clearer understanding of our experimental procedures but

also ensures the replicability of our findings. [Section 6.2](#) analyzed the benchmark and verified the efficiency of the proposed bi-level model. [Section 6.3](#) conducted the numerical analyses in terms of the computational performance and solution quality of the proposed solution approach.

6.1. Test instances

In this research, we meticulously analysed the comprehensive operational dataset of a specific aircraft model A319 obtained from a prominent Chinese airline over one week. This dataset was chosen for its relevance and completeness. The airline operates a fleet of 22 aircraft of this specific model, which service 20 airports and conduct 643 flights during the observed period. Note that, here we only consider two maintenance stations that have the corresponding maintenance qualifications for the specific aircraft model.

To conduct a systematic experimental analysis, we divided the row dataset into ten distinct computational instances. These instances are presented in [Table 4](#), where each instance encapsulates the critical operational parameters, which include the number of aircraft involved, the total number of flights undertaken, and the number of airports the fleet serves.

Table 4. Summary of Case Scenario.

Case Scenario	# Aircraft	# Flight	# Airport	# Maintenance Station
S1	3	86	20	2
S2	5	145	20	2
M1	7	193	20	2
M2	9	244	20	2
M3	11	299	20	2
L1	13	354	20	2
L2	15	408	20	2
L3	17	462	20	2
L4	19	535	20	2
L5	22	643	20	2

Additionally, we provide comprehensive financial parameters such as the operating cost per hour and the maintenance cost per flight hour. As delineated in [Table 5](#), the data from the International Air Transport Association (IATA)⁶ indicate that the maintenance cost per FH for this aircraft model is \$1,340. Line maintenance accounts for 24% of these expenditures, equal to a specific cost of \$321 per FH. According to operational data provided by the [Civil Aviation Administration of China \(CAAC\)](#),⁷ these aircraft's average hourly operating cost is estimated to be \$6,076 per FH.

Table 5. Description of Some Parameters.

Parameter	Description	Value
cf	Flight cost per hour (\$/FH)	6076
cm	Maintenance cost per flight hour (\$/FH)	321
α	Flexible maintenance cost ratio	1.5
N	Absolutely large constant	1,000,000
H_{max}	Maximum flight hours	65
F_{max}	Maximum number of takeoffs	20

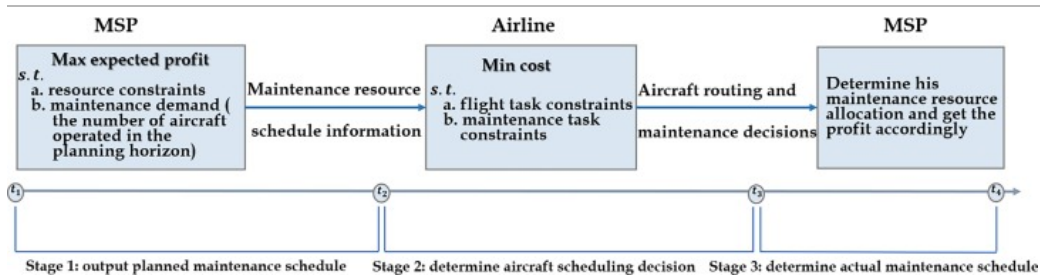
[Table 6](#) shows computational rules for flight operating cost of each flight and the maintenance cost of each aircraft. It is pivotal to make theoretical models align with real-world operational costs and maintenance expenses. By incorporating these financial parameters into our analysis, it provides a more realistic and economical assessment of aircraft operations.

Table 6. Relationship between Parameters.

Parameter	Value
$c_{j,t}$	$c_f \times$ Flight f^t 's operating time.
c_{jmt}	$cm \times$ Accumulated flight hours for maintenance at station m on day t .

6.2. Benchmark

To validate the effectiveness of the proposed bilevel model, we build a benchmark model based on existing literature in which there is no information sharing between the airline and MSP. The decision mechanism of this benchmark model is based on a sequential decision process. The decision sequence and relationships are illustrated in Fig. 4. First, at the beginning of the decision stage, the MSP does not have specific aircraft maintenance plans from the airline. He estimates aircraft maintenance demands based on the number of aircraft operated by the airline in the coming operational period. Considering its maintenance resource constraints (including skilled maintenance man-hours and aircraft hangar space in each maintenance station), the MSP maximizes its revenue for the current decision cycle. In this scenario, the MSP obtains the optimal maintenance resource schedule and provides it to the airline. In the second stage, the airline minimizes its operating costs for the current decision cycle based on the maintenance resource schedule provided by the MSP and its flight schedules and then provides feedback on the aircraft paths and maintenance task arrangements to the MSP. In the third stage, the MSP determines the specific maintenance resource allocation and scheduling schemes based on the airline's maintenance requirements and obtains its actual revenue.



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Fig. 4. The Decision Stages in the Benchmark Model.

Table 7 shows the outcomes of the benchmark model. The first three columns respectively represent the RA objective values, the LP value of the AMRP, and the IP value of the AMRP under the benchmark. The AMRP's Gap is $(AMR - PIP - AMRP - LP) / AMRP - LP$. The computational time for AMRP, CG-Time, is recorded to analyze computational complexity. It is noted that the runtime exhibits a compound growth pattern with the increasing problem scale. The number of iterations in the column generation process is indicated by # Iteration. # Branch refers to the count of branching operations performed to achieve integer solutions. Furthermore, to analyze the complexity of the model, the numbers of rows and nonzeros are also recorded, denoted as # Row and # Nonzero, respectively.

Table 7. Basic Experimental Results.

Instance	RA-Value	AMRP-LP	AMRP-IP	GAP	CG-Time	#Iteration	#Branch	#Row	#Nonzero
S1	9645.2	1482190.80	1482190.80	0%	4.99	48	0	131	2911
S2	7011.45	2399818.59	2406052.55	0.26%	24.42	70	2	192	6943
M1	7992.95	3193486.87	3220319.95	0.83%	152.55	124	3	242	16,314
M2	11867.06	4098481.08	4118990.45	0.50%	224.84	136	3	274	21,444
M3	16174.35	5099564.15	5132447.95	0.64%	610.05	148	6	331	31,517
L1	18252.7	6195316.23	6273686.45	1.25%	912.79	134	9	391	36,644
L2	20723.2	7021682.25	7295613.23	3.75%	1795.8	147	14	447	43,787
L3	33724.33	7894138.73	8380353.12	5.80%	3561.52	165	15	503	58,312
L4	40298.6	9395780.63	9898866.23	5.08%	5640.56	154	15	578	62,533
L5	58298.6	10831691.86	11365888.6	4.70%	16747.5	229	20	689	100,977

Note: All time units are in second.

6.3. Experimental results

6.3.1. Experimental results of the bilevel model

Table 8 presents the results obtained using the GSNCG algorithm for the bilevel model as proposed in Section 5. Here, Upper-obj represents the value of the objective function of the upper-level MSP. Lower-obj (LP) and Lower-obj (IP) display the optimal values of the solutions under linear relaxation and mixed integer programming models. The results show that all GAP values are within 5%, which is satisfactory as the linear relaxation is the lower bound for the optimal solution. Column # Bilevel Iteration represents the number of iterations in conducting the bilevel model, which equals the number of resource configuration combinations searched based on the upper-level model for the MSP. The final column in the table records the computational time to solve the proposed bi-level model. Although the computational time is relatively longer than the traditional single-level model, for the large problem scale, it takes less than 14h to get the optimal or near-optimal solution which is still acceptable for the weekly planning horizon.

Table 8. Bilevel Experimental Results.

Instance	Upper-obj	Lower-obj (LP)	Lower-obj (IP)	GAP	#Bilevel Iteration	Bilevel Run-time
S1	9673.70	1468809.30	1468809.30	0.00%	20	15.23
S2	15560.20	2335116.30	2335116.30	0.00%	41	160.45
M1	20671.35	3212771.75	3213735.58	0.03%	70	1086.00
M2	39170.63	3975694.03	3981666.53	0.15%	60	2422.91
M3	31612.70	4832170.89	4822044.60	0.21%	67	6931.72
L1	52954.37	6000104.37	6011526.27	0.19%	33	7620.57
L2	55762.82	7117939.14	7197107.32	1.10%	35	11648.67
L3	72787.72	8014410.32	8203081.18	2.30%	36	13072.33
L4	74959.42	9420201.54	9583114.48	1.70%	36	24259.14
L5	83782.82	10759360.0	11242800.42	4.30%	36	50132.39

Note: $GAP = \frac{IP-LP}{LP} \times 100\%$.

Table 9 presents a comprehensive evaluation of the efficiency of the proposed bilevel algorithm. The evaluation metrics include the total runtime for the bilevel model, the number of iterations of the bilevel model, the runtime for the initial column generation process, and the time required for a single execution of the label-setting algorithm to find an optimal path during the initial column generation. It is worth noting that the label-setting algorithm is invoked repeatedly during the column generation procedure, and its cumulative runtime is generally comparable to the overall runtime of the column generation.

Table 9. Analysis of Run Time for the Bilevel Model.

Instance	Bilevel Run-time	#Bilevel Iteration	CG-time	Labeling -time
S1	15.23	20	3.76	0.12
S2	160.45	41	30.55	0.31
M1	1086.00	70	150.00	1.20
M2	2422.91	60	229.00	1.80
M3	6931.72	67	899.82	4.80
L1	7620.57	33	1123.00	7.80
L2	11648.67	35	1512.33	14.50



New

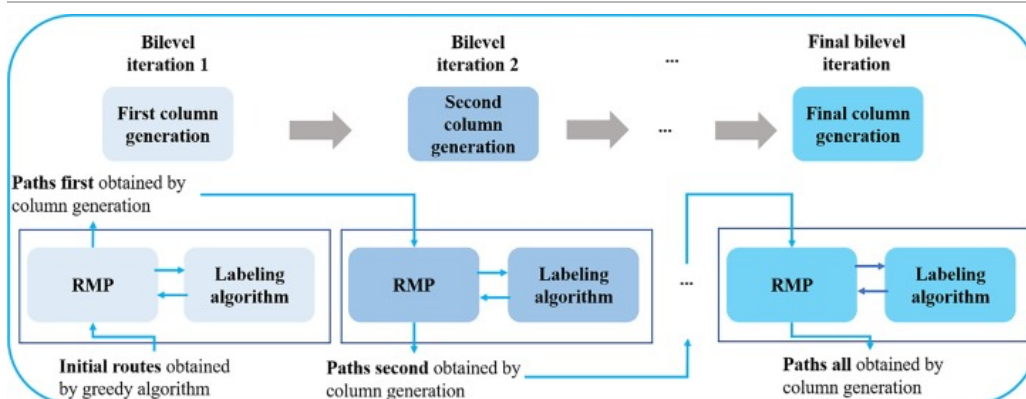
Instance	Bilevel Run-time	#Bilevel Iteration	CG- time	Labeling -time
L3	13072.33	36	3077.67	30.40
L4	24259.14	36	7344.77	48.00
L5	50132.39	36	16548.12	82.00

It is revealed that the total runtime of the bilevel model is much less than the product of the runtime for the column generation procedure and the number of bilevel iterations. This finding strongly verifies the efficiency of our proposed algorithm. Additionally, the runtime of the labeling algorithm is increased as the size of the instance increases. This, in turn, affects the total runtime of the bi-level model. As each bilevel iteration requires the execution of the lower-level AMRP column generation algorithm, it requires multiple calls to the labeling algorithm.

To understand the efficiency improvements of our proposed bilevel model algorithm, we compare the number of paths generated during the first iteration of the column generation process (“# Paths First”) with the total number of paths generated throughout the entire bilevel programming process (“# Paths All”), as indicated in Table 10. It is found that the total number of paths at the end of the bilevel iterations is not as large as initially expected. It is, in fact, quite similar to the number of paths generated during the first iteration, rather than scaling with the product of the first iteration’s paths and the total number of bilevel iterations. This is because, in our approach, each iteration retains paths from the previous iteration, which reduces the need to invoke the time-intensive label-setting algorithm repeatedly during column generation, as shown in Fig. 5. Table 11 shows that this adaptive path retention mechanism significantly reduces both the number of paths generated and the number of label-setting calls, resulting in reduction on redundant path generation and enhanced computational efficiency.

Table 10. Analysis of the Paths Generated by the Bilevel Model.

Instance	# Paths All	# Bilevel Iteration	# Paths First
S1	505	20	217
S2	1252	41	508
M1	4507	70	1053
M2	5583	60	1369
M3	6592	67	1929
L1	5619	33	2939
L2	4809	35	3264
L3	3711	36	3526
L4	5295	36	4809
L5	5896	36	5762



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Fig. 5. Illustration for Efficient Path Record Mechanism.

Table 11. Cost Reduction for the Airline Company.

Instance	AMRP-LP	AMRP-IP	Lower-obj (LP)	Lower-obj (IP)	Cost Reduction	Improvement Percentage
S1	1482190.80	1482190.80	1468809.30	1468809.30	13381.50	0.90%
S2	2399818.59	2406052.55	2335116.30	2335116.30	70936.25	2.95%
M1	3193486.87	3220319.95	3212771.75	3213735.58	6584.37	0.20%
M2	4098481.08	4118990.45	3975694.03	3981666.53	137323.92	3.33%
M3	5099564.15	5132447.95	4832170.89	4822044.60	310403.35	6.05%
L1	6195316.23	6273686.45	6000104.37	6011526.27	262160.18	4.18%
L2	7021682.25	7295613.23	7117939.14	7197107.32	98505.91	1.35%
L3	7894138.73	8380353.12	8014410.32	8203081.18	177271.94	2.12%
L4	9395780.63	9898866.23	9420201.54	9583114.48	315751.75	3.19%
L5	10831691.86	11365888.6	10759360.0	11242800.4	123088.18	1.08%

6.3.2. Cost reduction and profit enhancement: Benefits for the airline and maintenance service provider

To evaluate the economic benefits of the bilevel model for the airline and the MSP, we conduct a comparative analysis of financial indicators under the bilevel model versus the benchmark. Specifically, we compare the total profits for the MSP and the total costs for the airline under these two models.

Table 11 displays the cost reduction of the airline when implementing the interactive decision-making structure compared to the benchmark among all the instances. The columns “Lower-obj (LP)” and “Lower-obj (IP)” accurately record the linear relaxation and integer optimal values under the bilevel decision making structure, respectively. The results indicate that the bilevel model significantly outperforms the benchmark regarding cost control. The extent of cost reduction varies across different instances, which shows less correlation with the problem size. It is indicated that the cost reductions are consistently positive and significant across all instances under the proposed bi-level model.

Similarly, Table 12 summarizes the comparison result for the MSP when applying the interactive decision-making structure. Compared to the solution of the benchmark (i.e., RA-value), it is obvious that there is always a positive profit improvement for each instance. And it is verified that there is a significant profit improvement achieved for all the instances (except S1) by the proposed bi-level model no matter the size of the problem it is.

Table 12. Profit Improvement for Maintenance Provider.

Instance	RA- Value	Upper- obj	Profit Increase	Improvement Percentage
S1	9645.20	9673.70	28.50	0.30%
S2	7011.45	15560.20	8548.75	121.55%
M1	7992.95	20671.35	12678.40	158.62%
M2	11867.06	39170.63	27303.57	230.08%
M3	16174.35	31612.70	15438.35	95.45%
L1	18252.70	52954.37	34701.67	190.12%
L2	20723.20	55762.82	35039.62	169.08%
L3	33724.33	72787.72	39063.39	115.83%
L4	40298.60	74959.42	34660.82	86.01%
L5	58298.60	83782.82	25484.22	43.70%

In summary, the interactive decision making structure reflected in the bilevel model not only boosts the MSP's profit but also reduces the total operating costs for the airline. This outcome stems from effective information exchange between both parties through the proposed interactive decision making structure, which leads to a win-win situation, compared to the traditional sequential decision making model.

6.3.3. Comparison between the proposed algorithm and the existing algorithm

In this section, to further validate the performance of our algorithm in terms of solution quality and computational efficiency, a comparison is conducted with the heuristic algorithm. Referring to the existing literature ([Camacho-Vallejo et al., 2024](#), [Sinha et al., 2017](#)), the nested Ant Colony Optimization (ACO) algorithm is a mainstream *meta*-heuristic algorithm for solving bilevel programming problems. Therefore, ACO is selected as the comparative algorithm for in-depth analysis. The proposed algorithm is compared with the nested ant colony iteration mechanism proposed by [Eltoukhy et al. \(2018\)](#), and the results are shown in [Table 13](#).

Table 13. Comparison between the Proposed Algorithm and ACO Algorithm.

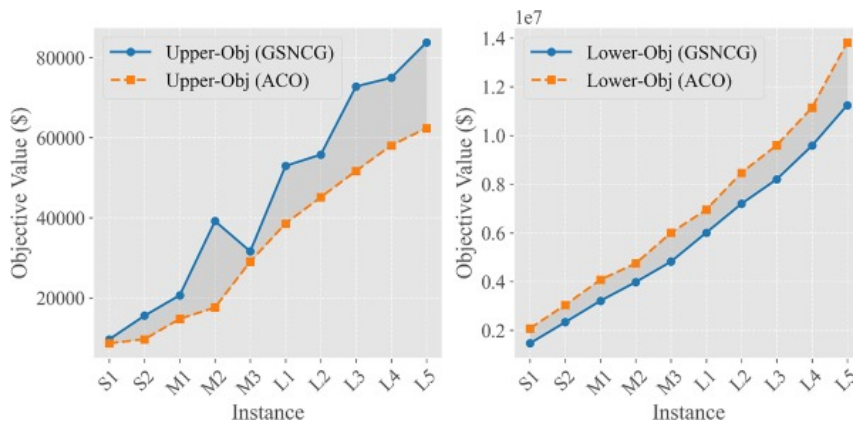
Instance	Upper-Obj	Upper-Obj (ACO)	Upper-Obj GAP	Lower-obj (IP)	Lower-obj (IP)(ACO)	Lower-obj (IP) GAP	Bilevel Run-time	Bilevel Run-time(ACO)
S1	9673.7	8680.7	993	1468809.3	2079415.3	-610606.00	15.23	60.23
S2	15560.2	9690.25	5869.95	2335116.3	3043070.85	-707954.55	160.45	274.14
M1	20671.35	14763.75	5907.6	3213735.58	4,079,670	-865934.42	1086.00	1775.63
M2	39170.63	17615.5	21555.13	3981666.53	4748106.4	-766439.87	2422.91	2336.88
M3	31612.7	29071.75	2540.95	4822044.6	6006331.15	-1184286.55	6931.72	5460.9
L1	52954.37	38,568	14386.37	6011526.27	6954228.9	-942702.63	7620.57	6594
L2	55762.82	45129.25	10633.57	7197107.32	8471871.65	-1274764.33	11648.67	5807
L3	72787.72	51,681	21106.72	8203081.18	9600296.3	-1397215.12	13072.33	5694
L4	74959.42	58068.2	16891.22	9583114.48	11142580.1	-1559465.62	24259.14	11,389
L5	83782.82	62378.45	21404.37	11242800.4	13823141.6	-2580341.13	50132.39	17,508

Note: The Upper-Obj GAP = $Upper-Obj - Upper-Obj(ACO)$, and the Lower-obj (IP) GAP = $Lower-obj(IP) - Lower-obj(IP)(ACO)$.

As shown in [Table 13](#), it can be clearly observed that in terms of the upper-level objective values (Upper - Obj), our proposed algorithm generally obtains higher values compared to the ACO algorithm. The positive Upper-Obj GAP values indicate that our algorithm can achieve better solutions for the upper-level decision maker for all instances. For example, in instance S2, the Upper-Obj value of our algorithm is 15560.2, while that of ACO is 9690.25, resulting in a GAP of 5869.95, namely a percentage improvement of 60.6%. The negative Lower-Obj (IP) GAP values indicate that our algorithm can achieve better solutions for the lower-level decision maker for all instances. For example, in instance L5, the Lower-obj (IP) value obtained by the proposed algorithm is 11242800.4. In contrast, the IP value achieved by the Ant Colony Optimization (ACO) algorithm is 13823141.6, resulting in a GAP value of -2580341.13 with a percentage improvement of 18.7%. In addition, the difference between the two algorithms becomes more significant with the increase of the problem size. [Fig. 6](#) displays the performance of both the proposed algorithm and the ACO algorithm regarding the objective values.



New

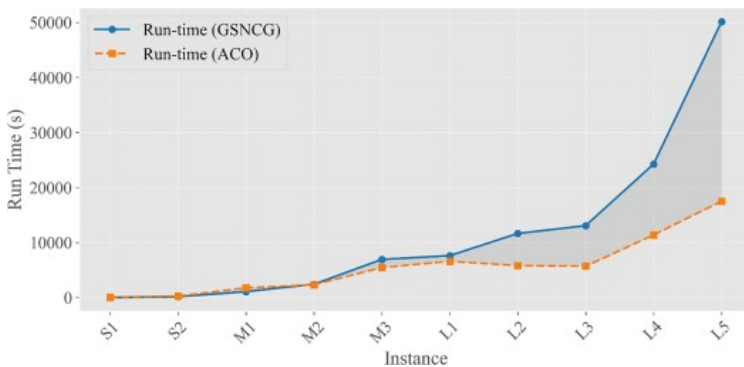


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Fig. 6. Comparison of the Objective Value between the GSNCG and ACO.

In terms of computational efficiency, as reflected by the Bilevel Run-time, for small-scale instances, our algorithm has a shorter run time. For example, for instance S1, the run time of our algorithm is 15.23s, while that of ACO is 60.23s. This indicates that our algorithm has advantages in solution quality and computational efficiency. However, as is shown in Fig. 7, when the problem size becomes larger, the computational times of both algorithms start to increase quickly. This is because, for large-size problems, it involves more iterations between the upper-level and lower-level models. We can also find that the ACO is much faster than the proposed exact method. In the future, it may be promising to consider integrating heuristics into the exact solution framework to further speed up the computational efficiency.



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Fig. 7. Comparison of the Run-time between the GSNCG and ACO.

7. Sensitivity analyses

In this section, we undertake sensitivity analyses in small and medium-scale instances to examine the effects of flexible maintenance costs ratio, opportunity cost, and the proportion of flexible maintenance resources. It provides a comprehensive understanding of the impacts of the flexible maintenance resources and opportunity cost on the bilevel optimal solutions, and some useful managerial insights are obtained as well.

7.1. Sensitivity analysis of flexible maintenance premium price ratio

This section explores the impact of flexible maintenance costs, specifically the premium price coefficient, on the optimal solutions. We conduct the sensitivity analyses on two small-scale instances (S1, S2) and three medium-scale instances (M1, M2, M3) to assess how the premium price coefficient α influences the results of the proposed bi-level model. We adjust the value of α , which represents the extra payment for flexible maintenance resources to simulate different scenarios. In Scenario One, we assume that the cost of flexible maintenance resources is equal to that of regular maintenance resources (*i.e.*, $\alpha = 1$), and treat it as the baseline for comparison across scenarios. Since acquiring flexible resources typically incurs higher costs, we also defined Scenario Two (*i.e.*, $\alpha = 1.5$) and Scenario Three (*i.e.*, $\alpha = 3$) to represent extra maintenance cost scenarios. The outcomes of these experiments can be seen in Table 14, which displays the variation in optimal solution values under different levels of premium price coefficient α . Notably, all cases listed in the table were solved within two hours.

Table 14. Sensitivity Analysis-Flexible Maintenance Premium Price Ratio.

Instance	α	Upper-obj	Lower-obj (IP)	Bilevel Run-time	RA Gap	AMRP Gap
S1	1	9663.20	1456238.80	19	baseline	baseline
S1	1.5	9673.70	1468809.30	15	0.11%	-1.37%
S1	3	11063.20	1477632.04	21	14.49%	-1.98%
S2	1	12860.83	2264368.93	257	baseline	baseline
S2	1.5	15560.20	2335116.30	237	20.99%	-3.12%
S2	3	15872.20	2396895.20	240	23.42%	-5.85%
M1	1	18009.95	2986756.95	1134	baseline	baseline
M1	1.5	20671.35	3213735.58	1086	14.78%	-7.60%
M1	3	21547.95	3429370.45	1173	19.64%	-14.82%
M2	1	34644.45	3968112.77	2643	baseline	baseline
M2	1.5	39170.63	3981666.53	2422	13.06%	-0.34%
M2	3	31353.45	4139449.45	2235	-9.50%	-4.32%
M3	1	28110.95	4793382.95	5023	baseline	baseline
M3	1.5	31612.70	4822044.60	4932	12.46%	-0.60%
M3	3	36241.53	5213015.53	5374	28.92%	-8.75%

Note: the RA Gap = $\frac{\text{Upper_obj} - \text{Upper_obj}(\text{baseline})}{\text{Upper_obj}(\text{baseline})} \times 100\%$, and the AMRP Gap = $\frac{\text{Lower_obj} - \text{Lower_obj}(\text{baseline})}{\text{Lower_obj}(\text{baseline})} \times 100\%$.

Here, the “RA Gap” column denotes the percentage increase in profits for the MSP compared to the baseline scenario of the sensitivity analysis. The “AMRP Gap” indicates the percentage reduction in the operating costs compared to the baseline scenario; a negative percentage indicates an increase in the operating cost to the baseline scenario, whereas a positive percentage indicates a decrease in the operating cost. It is revealed that the total operating cost of the airline always increases as the extra payment coefficient increases (as shown in the fourth and last columns). For the MSP, the situation is generally the opposite, with the profits mostly showing an upward trend as α increases (eg., S1, S2, M1, M4). However, there is an exception where the MSP's profit increases and then decreases along with α , which is even below the baseline scenario (see M2). The reason behind that is, the airline changes its aircraft routing decisions. The extremely high maintenance costs force the airline to cancel some of the flights to avoid the originally necessary maintenance tasks for some aircraft during the current planning horizon. Thus, the MSP may lose profit due to fewer maintenance tasks being performed.

Managerial Insights: When the MSP gains more pricing power over flexible maintenance resources, it challenges airline operational cost management. The MSP must price these resources carefully based on airline's demands, as simply raising prices isn't always profitable and could harm both parties. Analyzing the flexible maintenance cost coefficient shows that an excessive price hike by the MSP might boost short-term profit but risk revenue loss if airlines cut or postpone maintenance. Thus, in practice, the MSP should consider airlines' reliability and cost sensitivity, control price increases, and use strategies like rolling adjustments or phased quotations considering factors like seasonal demand and fleet size changes. The airline, when facing high premiums, should prioritize maintenance for key routes or aircraft to avoid larger operating losses from flight cancellations or delays due to over economizing on maintenance.

7.2. Sensitivity analysis of opportunity cost

In this section, we conduct the analysis based on the first five instances, where the opportunity cost C_F is adjusted to evaluate its impact on the bilevel optimal solution. Specifically, we establish four scenarios: in Scenario One, the MSP solely focuses on his profit without considering the operational status of the cooperative airline (*i. e.*, $C_F = 0$); in the other three scenarios (*i. e.*, $C_F = 100, 200, 300$), the MSP takes into account the operational situation of the airline's flights and the value of C_F reflects the cooperative relationship between the two players. A higher C_F refers to a higher negotiation power from the lower-level player (*i. e.*, the airline). Among them, Scenario Two is considered the baseline scenario for the sensitivity analysis. Table 15 displays the optimal bilevel solutions under different C_F values.

Table 15. Sensitivity Analysis – Opportunity Cost.

Instance	C_F	Upper-obj	Lower-obj (IP)	Bilevel Run-time	RA Gap	AMRP Gap
S1	100	9673.70	1468809.30	15	baseline	baseline
S1	200	7663.20	1449750.80	22	-20.78%	1.30%
S1	300	5673.70	1447723.30	18	-41.35%	1.44%
S1	0	11663.20	1472458.80	39	20.57%	-0.25%
S2	100	15560.20	2335116.30	237	baseline	baseline
S2	200	11860.20	2333827.05	225	-23.78%	0.06%
S2	300	9860.83	2276526.93	246	-36.63%	2.51%
S2	0	19260.20	2380502.88	232	23.78%	-1.94%
M1	100	20671.35	3213735.58	1086	baseline	baseline
M1	200	15525.45	3180086.95	1247	-24.89%	1.05%
M1	300	8721.95	3153934.95	1425	-57.81%	1.86%
M1	0	24052.70	3296938.70	1157	16.36%	-2.59%
M2	100	39170.63	3981666.53	2422	baseline	baseline
M2	200	35919.45	3875065.95	2246	-8.30%	2.68%
M2	300	28080.7	3813380.10	2271	-28.31%	4.23%
M2	0	40090.7	4156660.20	2308	2.35%	-4.39%
M3	100	31612.70	4822044.60	4932	baseline	baseline
M3	200	30348.95	4766445.85	5032	-4.00%	1.15%
M3	300	28333.78	4748361.42	4675	-10.37%	1.53%
M3	0	38063.78	4958737.97	5012	20.41%	-2.83%

It is shown that, as C_F increases, the operating cost of the airline is reduced, while the maintenance servicer provider tends to lose his profit. In other words, when the upper-level player (i.e., the service provider) considers a higher opportunity cost in his objective, it brings benefit for the lower-level player (i.e., the airline). However, it may have negative impacts on the upper-level player. But it is interesting to note that, even under Scenario One (i.e., self-interest-oriented), the obtained bilevel optimal solutions are better than the optimal solutions under the sequential decision-making process.

Managerial Insights: Considering the opportunity cost of the lower-level party in the upper-level decision-making process reflects the negotiation power between two parties, which benefits more for the lower-level party. When determining the opportunity cost coefficient, both parties need to balance long-term cooperation and short-term interests. Appropriately setting the opportunity cost (e.g., C_F between 0 and 100) helps reconcile short-term interests for a win-win situation, promoting long-term cooperation. The results on opportunity cost show that when the MSP values the airline's operating conditions more in its decision-making (higher opportunity cost coefficient), it can reduce the airline's costs but cut its own short-term profits. This indicates that a win-win outcome often stems from long-term cooperation. For the MSP, setting the opportunity cost coefficient appropriately helps build a closer partnership and stable revenue. It can include "cooperation depth" or "service quality guarantee" clauses in contracts with airlines, sacrificing some premium for long-term maintenance business. For the airline, leveraging its core position in capacity and route planning can provide compelling operating data during negotiations to enhance the MSP's willingness to cooperate and increase concession space.

7.3. Sensitivity analysis of maintenance resources

To further analyze the impacts of the available maintenance resources on the optimal decisions of the two players, we assume three different scenarios. Scenario one: The MSP only offers very few additional maintenance service chances to the airline company (i.e., $N = 1$). Scenario two: the MSP offers a certain number of additional maintenance service chances to the airline company (i.e., $N = 3$). Scenario three: the service provider offers enough additional maintenance service chances to the airline

company (*i.e.* $\mathfrak{N} = 6$). Here, $\mathfrak{N}$ is defined as the difference between the number of maintenance service chances accessible to the airline and the maximum number of aircraft needed to conduct maintenance tasks during the planning horizon for the airline.

As shown in Table 16, for the MSP, it is always better to offer more maintenance chances, including regular and flexible maintenance resources, which leads to higher profits. Moreover, the more resources it is provided, the greater its profit is. However, for the airline company, the situation is relatively complex. It can be seen that as the MSP increases the chances to conduct maintenance tasks, the operating cost of the airline keeps decreasing in some instances (e.g. S1, M2), while in other cases, it initially decreases and then increases (e.g. S2, M1, M3). However, in most cases, having more maintenance service chances accessible by the service provider will allow the airline to have more opportunities to optimize the aircraft maintenance routing decisions, thereby further reducing the total operating costs. However, as the upper-level decision-maker, the service provider tends to offer expensive flexible maintenance resources to increase its profits. This undoubtedly forces the airline to incur higher maintenance costs and may further affect its aircraft routing decisions. **Therefore, it can be seen that when the MSP dominates the decision-making process, providing more heterogeneous maintenance service chances does not always bring better choices for the airline company.**

Table 16. Sensitivity Analysis of Maintenance Resources.

Instance	$\mathfrak{N}$	Upper-obj	Lower-obj (IP)	Bilevel Iteration	Bilevel Run-time	RA Gap	AMRP Gap
S1	1	9663.20	1482190.80	4	9	baseline	baseline
S1	3	9673.7	1468809.3	20	15	0.11%	0.90%
S1	6	9930.5	1444736.1	41	34	2.77%	2.59%
S2	1	6754.95	2399389.05	5	54	baseline	baseline
S2	3	15560.20	2335116.30	41	237	130.35%	2.68%
S2	6	16765.06	2385550.30	179	734	148.19%	0.58%
M1	1	20501.70	3295121.20	7	311	baseline	baseline
M1	3	20671.35	3213735.58	70	1086	0.83%	2.47%
M1	6	27008.45	3237080.95	363	13,250	31.74%	1.76%
M2	1	16840.20	4059407.20	6	356	baseline	baseline
M2	3	39170.63	3981666.53	60	2422	132.60%	1.92%
M2	6	58584.28	3951696.62	407	19,560	247.88%	2.65%
M3	1	28620.95	4975718.70	6	1119	baseline	baseline
M3	3	31612.70	4822044.60	67	4932	10.45%	3.09%
M3	6	46758.95	5014878.95	425	53,612	63.37%	-0.79%

It is indicated that when the total accessible maintenance resources are fixed, the ratio of the regular maintenance resources to the flexible maintenance resources may have essential impacts on the optimal decisions of the two players. To analyze the maintenance resources allocation strategy to achieve Pareto improvement, we assume there are three possible cases regarding the number of flexible maintenance resources being available, namely, high-level ($\mathfrak{L} = 1$), medium-level ($\mathfrak{L} = 2$), and low-level ($\mathfrak{L} = 3$). Here, $\mathfrak{L}$ is defined as the ratio of the regular maintenance resources to the flexible maintenance resources.

As is displayed in Table 17, if the MSP sets a lower level of flexible maintenance resources (*i.e.* $\mathfrak{L} = 2$ or 3), the airline can always make cost-effective decisions with a lower operating cost. However, this may not always be the case for the service provider. In most cases, the service provider's profit increases first increase and then decreases with $\mathfrak{L}$. But for some cases, it keeps decreasing (*i.e.* S1) or increasing (*i.e.*, M2) when $\mathfrak{L}$ is reduced. The results are reasonable. **Whether the service provider gets higher profit or not depends on the degree of the supply-demand matching between the airline and the service provider.** When the airline has maintenance service requirements but does not reach the urgent level, the airline will choose to postpone some of the maintenance tasks to avoid high maintenance costs.

Table 17. Sensitivity Analysis of the Ratio of the Regular Maintenance Resources to the Flexible Maintenance Resources.

Instance	$\square$	Upper-obj	Lower-obj (IP)	Bilevel Run-time	RA Gap	AMRP Gap
S1	1	9614.57	1476259.67	17	baseline	baseline
S1	2	5664.70	1459333.80	16	-41.08%	1.15%
S1	3	5664.70	1459333.80	18	-41.08%	1.15%
S2	1	15248.70	2422680.80	241	baseline	baseline
S2	2	15879.95	2369618.05	246	4.14%	2.19%
S2	3	12355.07	2349030.17	238	-18.98%	3.04%
M1	1	18937.98	3261734.48	1754	baseline	baseline
M1	2	24520.45	3203465.95	1931	29.48%	1.79%
M1	3	23919.65	3181719.15	1936	26.31%	2.45%
M2	1	33877.12	4021140.12	2003	baseline	baseline
M2	2	40202.12	3881288.02	2170	18.67%	3.48%
M2	3	48891.62	3823556.62	2095	44.32%	4.91%
M3	1	42962.57	5051099.57	5017	baseline	baseline
M3	2	54436.78	4834361.78	6013	26.71%	4.29%
M3	3	40406.20	4802984.70	5372	-5.95%	4.91%

However, when the service provider increases the regular maintenance resources (namely, the $\square$ increases from 1 to 2), the airline will reschedule its aircraft and more aircraft are planned to have maintenance service to avoid flight cancellation. In this case, the profit of the service provider increases. But, when the service provider keeps a much lower level of flexible maintenance resources (i.e., $\square$ is 3) and the airline can get access to most of the maintenance resources with the regular maintenance cost, the profit of the service provider does not increase but decreases. This is because the maintenance requirements from the airline are saturated, and no more maintenance requirements are needed. The service provider thus gets lost due to a lower maintenance price. However, the more regularly priced maintenance resources are available, the more chances for the airline to optimize its aircraft maintenance routing, further decreasing operating costs.

Managerial Insights: For the MSP, maintaining a certain level of flexible maintenance resources with higher prices can offer more initiative, yet an overly high level may prompt airlines to postpone maintenance. When facing diverse maintenance demands from multiple airlines, carefully determining the level of flexible maintenance resources $\square$ is crucial for achieving a win-win situation. Analyses of flexible maintenance resource availability ratios show that providing airlines with more or more flexible maintenance opportunities doesn't necessarily result in higher profit; It depends on the MSP's set maintenance fees as well as the airline's aircraft routing and maintenance plan adjustments. In reality, the MSP's blind increase in expensive flexible resources may lead the airline to compress or postpone planned maintenance, failing to achieve expected revenue growth. Thus, when strategically developing "multi-level" maintenance resources, the MSP must balance service quality and price elasticity, considering market demand and airline partner characteristics. The airline can utilize MSP-provided additional maintenance opportunities to boost on-time rates and operating efficiency for key flights or bottleneck routes, but it should avoid over-reliance on flexible resources that could drive up costs by intelligent aircraft maintenance planning. If both parties can negotiate a match in pricing and resource allocation, flights can operate smoothly, ensuring a win-win for both the MSP and the airline.

8. Conclusion

In this paper, we proposed a novel bilevel mixed integer programming model to study the mutually beneficial cooperative mechanism between the airline and the MSP from the perspective of the "supply-demand" system. Specifically, the MSP who faces the scarcity of resources is the upper-level decision maker, while the airline is the lower-level decision maker who aims to optimize its aircraft maintenance routing decisions. Different from the existing literature where the airline dominates negotiation power and ignores the resource scarcity of the service provider, the proposed bi-level model effectively characterizes the negotiation decision process between the MSP and the airline in real practice. In addition, we further propose an extra payment strategy for obtaining flexible maintenance resources. The airline can obtain flexible maintenance service by paying extra maintenance fees so that the reliability of the aircraft routing decision can be enhanced during real operations. The



numerical studies show that it not only reduces the total operating cost for the airline during aircraft routing but also increases the profit for the service provider. Therefore, the proposed bi-level model provides an effective theoretical modeling framework to realize the mutual benefit.

In our bilevel model, both upper and lower-level models are discrete optimization models which makes the problem more complex and difficult to solve for optimality. We first solved the lower-level AMRP model by reformulating the arc-based model into the path-based model, which can be further solved by using the column generation framework. In particular, to solve the difficulty in modeling multiple scenarios for each maintenance node, we simultaneously solve three sub-problems for generating different kinds of paths. For each kind of path generation, we use the labeling algorithm to find the shortest path for each scenario. To solve the bi-level model to optimality, we proposed an adaptive global search algorithm integrated with column generation. By recording the updated columns (i.e., paths) in each iteration, the computational time for the labeling algorithm is substantially decreased which improves the computational efficiency significantly. The computational experiments verified that our proposed solution approach is effective and efficient in obtaining optimal solutions.

Moreover, the sensitivity analyses regarding the flexible maintenance resources and opportunity cost reveal their impacts on the optimal solutions for both parties. **First**, when the MSP has increased pricing power over flexible maintenance resources, it poses a significant challenge to airline operational cost management. The service provider should carefully decide the price for flexible maintenance resources according to the demands of different airlines. It is not always profitable for the service provider to greatly increase the flexible maintenance service price. As it may lead to the outcomes that neither parties benefit. **Second**, taking into account the opportunity cost regarding the lower-level player in the upper-level decision-making process reflects the negotiation power of the lower-level player, which is more advantageous to the lower-level player. When both parties determine the opportunity cost coefficient, it is necessary to consider both the long-term cooperative relationship and the short-term interests of both parties. Appropriately setting the opportunity cost facilitates the tradeoff between the interests of both parties in a situation where both parties benefit, which promotes long-term sustainable cooperation. **Third**, when the MSP dominates the decision-making process, providing more heterogeneous maintenance service chances does not always bring better choices for the airline company. It depends on the degree of the supply–demand matching between the airline and the service provider. Therefore, as for the service provider, it is wiser to maintain a certain level of flexible maintenance resources with a higher maintenance price to give himself more initiative. However, an extremely high level of flexible maintenance resources will trigger the airline to postpone its maintenance requirements. When facing heterogeneous maintenance requirements from multiple airlines, carefully determining the level of flexible maintenance resources is critical for the service provider to achieve Pareto improvement with different airline companies.

The proposed exact bilevel decision-making framework offers several promising directions for future research. One crucial aspect is the incorporation of uncertainties, such as airline flight delays, into the model. This would enable the analysis of optimal strategies in uncertain environments. Additionally, the framework could be extended to a one-leader, multiple-follower structure, where a single MSP interacts with multiple airlines. This extension could utilize machine learning or reinforcement learning techniques to model more complex interactions among stakeholders, thereby enhancing both flexibility and scalability. Another valuable research direction lies in multi-objective optimization within the context of airline-MSP cooperation to make tradeoffs between efficiency, safety, and environmental protection. Finally, using data-driven methodologies for demand forecasting and decision-making is essential for real-world implementation. Such a decision support system should be capable of continuously updating model parameters based on real-time data, which would greatly enhance operational efficiency among different parties and render the model to achieve Pareto improvement under changing conditions.

CRedit authorship contribution statement

Xuting Sun: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xiaoyao Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Sai-Ho Chung:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Hoi-Lam Ma:** Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

The following are the Supplementary data to this article:

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Supplementary Data 1.

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